

Return on Investment for Promoting Homeownership: The case of the French Interest-Free Loan Policy

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Abstract

This article estimates the impact of subsidized loans for first-time homeownership on the number of new homeowners (extensive margin), housing choices (intensive margin), and housing prices (capitalization effect). Our identification strategy relies on the spatial and temporal variations of the French interest-free loan policy over the last decade, controlling for confounding treatment assignment through spatial semi-parametric propensity scores. Our doubly robust estimates cannot rule out that the policy has no effect at the extensive margin, while it has significant intensive margin and capitalization effects. For a wide range of social values for the extensive and intensive margins, we compute the returns to government spending according to counterfactual policy schemes and credit market conditions. Our simulations suggest that for reasonable social values, increasing public spending has a return on investment lower than one, and may even be negative in some situations.

JEL classification: H81, R21, R38

Keywords: Housing policy ; subsidized mortgage ; unconfoundedness ; generalized additive model ; spatial smoothing.

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1 Introduction

Access to homeownership is increasingly challenging in the context of the housing affordability crisis. The divergence between housing prices and household income represents the primary obstacle, not compensated by the decrease in interest rates in the 2010s (Barone et al., 2021). Not only does the difficulty in accessing homeownership raise concerns about inequality between households in terms of wealth accumulation (Sodini et al., 2023), but the slowdown in homeownership may also be detrimental from a social perspective. The positive externalities associated with homeownership, including improved housing maintenance (Harding, Miceli and Sirmans, 2000), school performance (Green and White, 1997; Harkness and Newman, 2003), or self-employment opportunities (Harding and Rosenthal, 2017) have been found to outweigh the negative ones, such as reduced residential mobility (Green and Hendershott, 2001), or risk of capital loss (Cunningham and Reed, 2013). Thus, homeowners generate positive externalities that amount to approximately \$1,300 per year (Coulson and Li, 2013). The slowdown of homeownership reduces these externalities, possibly leading to a loss for society. Hence, most developed countries support homeownership to increase welfare and tenants' aspirations.

The return on investment of public spending to support homeownership depends on its ability to increase homeownership, commonly defined as the extensive margin. Most policies leverage the financial channel to help renters qualify for a mortgage. Whereas the direct effect on tenure decision is not clear-cut, the stimulation of housing demand in the context of low elasticity of housing supply may inadvertently exacerbate affordability issues through price capitalization, and strengthen difficulties in access to homeownership. In addition to the effect on the extensive margin, helping renters qualify for a mortgage may also affect their housing choices (variation in size, location, or characteristics), commonly defined as the intensive margin. Although this effect is not necessarily negative from a political point of view, it does not encourage the production of positive externalities associated with homeownership. The most widely adopted policy supporting homeownership (the Mortgage Interest Deduction, MID) appears inefficient (Valentin, 2023) as it produces most of these effects in the intensive margin (Hanson, 2012), reinforcing affordability issues (Martin and Hanson, 2016) that are likely to be detrimental to the development of homeownership (Sommer and Sullivan, 2018). Hence, the policy return on investment is unclear as it depends on the balance between extensive and intensive margin effects regarding policy cost.

Several countries, such as France, have implemented alternative policy schemes to the MID, as recipients benefit from a complementary loan with no interest charge rather than income tax reduction. In these cases, policymakers determine both the proportion of the loan that will be subsidized (the *covering share*) and the maximum loan value (the *ceiling*

value). Although this type of policy has been implemented in various countries, policy evaluations are scarcer than MID in the literature with mixed results on the impact on homeownership (Gobillon and Le Blanc, 2008; Hembre, 2018; Carozzi, Hilber and Yu, 2024). Furthermore, these results are context-dependent¹, which makes the comparison difficult. In this paper, we estimate the return on investment for supporting homeownership development, jointly accounting for extensive and intensive margin effects, and price capitalization. We leverage the French Interest-Free-Loans (IFL) policy from 2015 to 2019² that provides financial support to potential first-time owners purchasing newly built housing units.³

We estimate the extensive and intensive margins responses to treatment intensity by leveraging spatio-temporal variations of the IFL policy at the French national scale. This French scheme employs a four-level spatial delineation, which defines the value of policy variables, i.e. the covering share and ceiling value for each year. This four-level design is called ABC zoning and aims to reflect local market tightness while being close to the affordability definition. Hence, spatiotemporal variations of treatment are endogenous to the municipality characteristics, since policy support is of greater importance in the most expensive municipalities. To handle this endogeneity, we exploit a selection-on-observables approach with a spatial smoothing specification to control for unobserved spatial confounders (Gilbert et al., 2024). While our approach is similar, in terms of identifying conditions, to a geographic regression discontinuity approach, we leverage all spatial heterogeneity in treatment intensity, which enables us to provide results that do not depend on local context.

We proceed as follows. We first estimate the probability for each municipality to receive each level of treatment from a large set of pre-treatment variables and a smooth bivariate function of geographical coordinates. The estimation procedure is carried out using an ordinal logistic regression accounting for the ordered structure of the treatment intensity and reflects, as in the policy scheme, that tightness is a latent variable. We then employ the Generalized Propensity Score (hereafter GPS, Imbens, 2000) as an inverse probability weighting mechanism to regress outcomes of the extensive and the intensive margins (respectively new homeowners and the housing choices of recipient) and capitalization effects (the average housing price). The assumption allowing us to interpret these results as policy effects is that, conditional on the set of pre-treatment variables and location, treatment variations are exogenous to municipality characteristics. We weigh observations

¹Hembre (2018) find a positive effect on the number of homeowners in the context of the Great Recession. In contrast, Carozzi, Hilber and Yu (2024) finds no significant effect for Greater London.

²This period contains three important reforms about treatment intensity leaving eligibility conditions unaffected. The policy also restricts access based on the income cap. Still, the income ceiling is not restrictive (Sotura, 2020).

³The French housing policy defined potential first-time owners as individuals who rent their main residence for at least two years.

according to their probability of belonging to another treatment level if the municipality had received that treatment level, using the GPS results. Then, we regress these reduced form policy effects on policy variables (the covering share and the ceiling value) and mortgage conditions (interest rate and loan maturity). It aims to disentangle the role of each policy variable in supporting homeownership, which is one of our main contributions. For extensive and intensive margin effects, we also account for the feedback effect resulting from the price capitalization. Finally, we compute the return on investment of additional public spending that accounts for policymaker objectives based on the marginal effect of one additional euro spent by policymakers.

The validity of our results depends on the specification of the treatment assignment. Our approach is four-fold to ensure good GPS specification. First, a large set of pre-treatment variables are collected, which are expected to relate to housing affordability. These variables encompass both demand-side (e.g. population density, income, socio-economic status) and supply-side (e.g. housing construction, past housing prices, past neighbourhood prices). Second, we account for spatial confounding variables (Gilbert et al., 2024) by introducing smoothing functions of the spatial coordinates of the location of the municipalities in a semiparametric Generalized Additive Model (hereafter GAM, Wood, 2017). For instance, geographical constraints (Saiz, 2010), housing supply elasticity (Accetturo et al., 2021), or demand for particular amenities (Bayer et al., 2016) might vary smoothly in space and likely to determine tightness measure. The only remaining threat for GPS misspecification is omitted variables with variations from one street to another such as school quality or air pollution due to traffic road being correlated with treatment intensity. Third, we use a doubly robust estimator to prevent potential misspecification in the GPS, including omitted variable bias. The variables in the first step are included in the outcome regression.⁴ The consistency of policy-relevant treatment effects can be ensured if, either the first, or second step of the identification strategy is well-defined (Robins and Rotnitzky, 1995; Słoczyński and Wooldridge, 2018). Finally, we compare our main results with alternative ones derived from a quasi-experimental approach. Leveraging the 2018 reform using the difference-in-difference approach, we find similar results to those obtained through our selection-on-observables approach. Nevertheless, our results are more general as they allow disentangling policy effects according to the source of variation (ceiling values, covering share and mortgage conditions) and overcoming common LATE issues (Imbens, 2010).

Our contribution to the literature is threefold. First, we estimate the efficiency of public spending to support homeownership accounting for both extensive and intensive margin effects. Although previous papers have estimated the policy effects at both margins, none of them considers these effects jointly in line with policymakers' objective through an

⁴(For applications of the doubly robust estimator see e.g. Brei et al., 2023; Deng et al., 2023; Hoang-Duc et al., 2024).

efficiency measure. In addition, we introduce a feedback effect to account for negative impacts on tenure decisions and housing choices resulting from price capitalization. We estimate that increasing public spending through the IFL is at best inefficient, as unintended effects are likely to outweigh targeted ones in credible scenarios. Second, we disentangle the impact of each policy variable on both extensive and intensive margin results. Considering that each policy variables have heterogeneous effects according to the population it affects, we estimate their marginal impacts on the extensive margin, the intensive margin, and price capitalization. While the ceiling value enhances windfall effects for housing choices (switching from nonsubsidized to subsidized segments), the covering share is mainly responsible for the capitalization effect. Third, we overcome common external validity issues using the selection-on-observables approach, making our results less context-dependent. Not only are our results in line with previous ones using natural experiments, but they also widen the external validity of the results, making policy results more applicable from an international perspective.

The remainder of the paper is structured as follows. [Section 2](#) presents the institutional context of the French IFL policy and the datasets we gather. In [Section 3](#), we introduce the return on investment function that accounts for the effects of the extensive and intensive margins. Our identification restriction to tackle endogenous treatment intensity and the doubly robust estimation procedure are presented in [Section 4](#). [Section 5](#) provides the empirical results from the two-step approach and [Section 6](#) concludes.

2 Institutional Context and Data

2.1 The French IFL Policy

The IFL policy was introduced in 1995 in France to encourage first-time homeownership. Recipients benefit from a complementary loan being publicly supported with no interest to pay for a given share of their overall housing credit (the covering share) up to a maximum value (the ceiling value). The cost for public finance amounts to the interests not charged for recipients of the IFL, as the government compensates bank losses through tax deductions. The total number of recipients since the beginning of the policy is approximately 3.2 million, for an overall budget cost of about 26.1 billion euros (about 8,000 euros per recipient). The policy excludes a small share of high-income households (about 10% of the richest tenants according to Sotura, [2020](#)). Specifically, it targets the newly-built housing units (although existing housing is eligible conditionally for renovation from 2016). Hence, all first-time owners cannot benefit from public support in the transition to homeownership, especially those purchasing existing housing units. We account for potential opportunistic behaviour of potential first-time owners switching from non-subsidized

to subsidized housing as a result of the policy.

Any IFL contract must be associated with a classical loan with interests to pay, so recipients should comply with the usual conditions to access the credit market.⁵ Hence, the total borrowed amount is the sum of a classic loan and the IFL one. Modifications to policy variables are made at the discretion of the policymakers every year. Yet, recipients cannot contract a new IFL if conditions are more or less favourable after the policy reforms, meaning that the IFL conditions are those in place at the time of the purchase. Similarly, recipients lose benefits from the policy if they sell their house to achieve residential mobility as they must reimburse the remaining IFL capital.

The four key parameters that shape the IFL intensity have remained stable over time. For each period and each group of municipalities (to which we will return below), policymakers fix s , ranging from 10% to 40%, the maximum coverage that the IFL can represent in the overall borrowed amount $\tilde{V}$ and $\bar{V}$, ranging from 100 to 150 thousand euros for one-person household, a maximum ceiling value on the loan. These are the two parameters at the government's disposal to implement the IFL policy, considering that the income cap for eligibility is not a binding condition. These two variables are not expected to have the same impact on the subsidy with a potential differentiated effect on extensive and intensive margins. On the one hand, increasing the covering share increases the subsidy for all policy recipients, regardless of the amount they borrow (left panel, Fig. 1). On the other hand, increasing the ceiling value increases the subsidy for the most expensive housing units, while leaving the least expensive ones unaffected (left panel, Fig. 1).⁶

$$c = \left[\frac{m \times r}{1 - (1 + r)^{-m}} - 1 \right] \times s \times \min(\tilde{V}, \bar{V}). \quad (1)$$

The budget cost of an IFL contract weakly increases with the four previous parameters, which indicates that increasing one of them is equivalent to increasing the financial support for recipients. This defines the intensity of treatment of the policy as an increase in one of the four considered parameters or an increase in the budget cost for the government. We restrict our studied period to the last three IFL reforms of the 2015–2019 period, as eligibility conditions remain unaffected and the ABC classification of municipalities (i.e. the zoning for the two parameters under policy control) does not change.

⁵For the French market, a common condition is that reimbursement payments cannot be higher than one-third of income. Consequently, this condition also restricts access to the IFL policy.

⁶More precisely, it affects home purchases that are censored, i.e. home purchases with purchase price higher than the previous ceiling value (from a reform perspective).

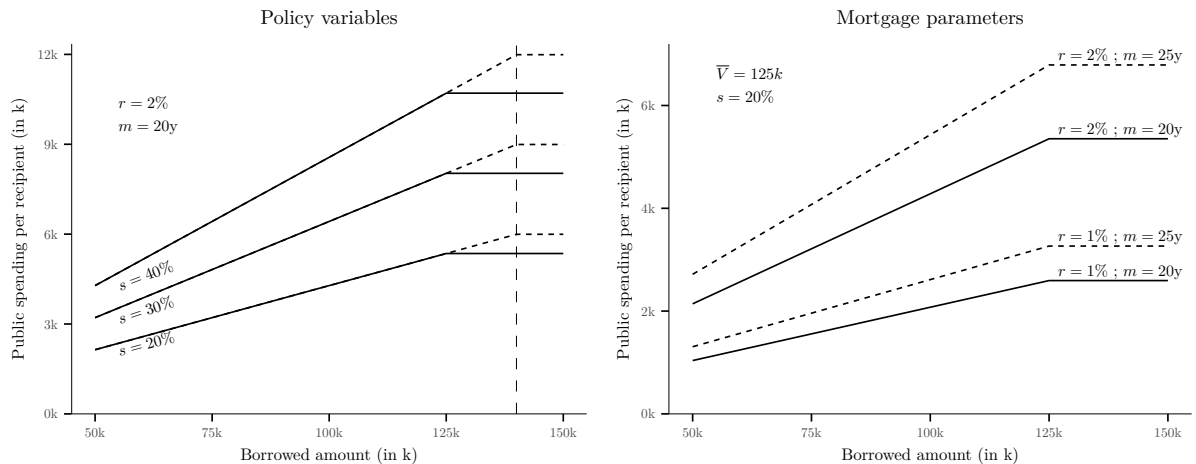


Figure 1. Cost of homeownership according to main policy variables

Notes: We simulate the cost of homeownership per recipient from the policymaker’s perspective according to the four policy variables. The left panel represents cost variation according to parameters controlled by policymakers (covering share and ceiling value), while the right panel consider variations in mortgage conditions (interest rates and maturity). The x-axis corresponds to the overall amount being borrowed by households, whereas the y-axis is the cost for the administration per recipient (and hence, the gain for purchasers). The overall budget cost also depends on the number of recipients.

2.2 The ABC Zoning

The two policy variables (covering share and ceiling value) depend on the location of IFL contracts, according to an exhaustive and mutually exclusive classification of French municipalities based on the housing market tightness. The municipality is the smallest jurisdiction for France with 34,970 municipalities.⁷ The objective of this spatial targeting is to rationalize public expenditures and provide more intense support to the most tightened areas, that is, the less affordable markets.⁸ Therefore, the policy variables defining the intensity of treatment are determined by the ABC treatment zone. The practical implementation of the tightness measure lies on four levels of zoning, from C the lowest level to A the highest level, with B₂ and B₁ as intermediate levels. This official zoning was updated four times since its introduction in 2003, the latest update of October 2014 is stable for the 2015–2019 period under study. Most French municipalities are rural and belong to zone C, the lowest level of the zoning (Panel A of Table 1). Table 1 also shows that the distribution of the ABC classification is consistent with general expectations, as municipalities with higher population densities and higher housing prices per living area (unitary prices hereafter) are higher in the housing tightness hierarchy.

Despite strong correlations between the ABC hierarchy and reported pre-treatment variables, the practical implementation of the tightness measure lacks transparency (Cour des Comptes, 2012). The zoning does not depend exclusively on objective character-

⁷Half of French municipalities have less than 457 inhabitants in 2019.

⁸According to official documents, tightness is defined as “the imbalance between the housing supply and the housing demand” (French Ministry of Ecological Transition).

istics, suggesting potential subjectivity in the assignment such as political discussion or variability in criteria. Hence, similar municipalities according to their real tightness measure (which is a latent variable) might be assigned to two different tiers. Most existing quasi-experimental approaches dealing with the endogeneity of the French housing policy assignment rely on the underlying errors of this zoning to recover policy effects (Labonne and Welter-Nicol, 2015; Beaubrun-Diant and Maury, 2021; Chareyron, Ly and Trouvé-Sargison, 2021). Our identification method leverages the underlying errors of the ABC zoning using a selection-on-observables approach to estimate policy effects.

2.3 Data

We aggregate three exhaustive individual data sources⁹ at the municipal level ($N = 34,970$) and match them with demographic data. We remove municipalities on the Corsican island due to the absence of spatial contiguity¹⁰ (360 observations) and municipalities of the *Alsace-Moselle* region (1,605 observations) as housing price data are missing for administrative reasons. Furthermore, we filter missing observations or data inconsistencies to obtain a final sample of 26,819 municipalities.¹¹ Most observations with missing variables concern median income in line with the French secrecy rule that imposes to have at least 11 observations to provide statistical information. These observations belong mainly to the C-tier, which is significantly different from the B₂ municipalities in the observable variables.¹² Consequently, potential selection bias is at worst marginal, as these observations have little impact on causal estimation considering their low probability of belonging to another tier.

IFL data. The first exhaustive database (provided by the SGFGAS) concerns all recipients subsidized by the IFL policy. Each recipient is located at the municipality level of its new home, with variables informing the loan contract (total value of the main and subsidized loans, total subsidies, interest rates, and maturity). These data also contain household characteristics such as annual income, number of members, matrimonial

⁹The data sources to which we have access are either at the recipient level (IFL database), the transaction level (the transaction database) or the housing unit for first-time owners (the fiscal database). However, we cannot match these data due to the lack of common identifiers and differences in spatial granularity.

¹⁰The introduction of spatial smoothing functions in the specification of the propensity score imposes that the probability is spatially continuous. However, geographical constraints such as the Mediterranean Sea violate these assumptions.

¹¹We report descriptive statistics in [Appendix A.2](#) for the relevant variables used in the empirical analysis

¹²Considering a selection model within municipalities C with binary variable as dependent one that indicates whether variables are filled, we find that municipalities with missing variables on median income have lower density and lower housing prices. Thus, they are likely to have a low weight in bilateral estimations as they have a low probability of belonging to another group.

status, and previous location when tenants. Finally, these data include characteristics of the housing concerned by the loan, such as construction date, surface, purchase price, and purchase date. We use them to construct aggregated values for each municipality, by computing for each year the number of IFL contracts and averaging loan, housing and household characteristics.

Tax data. To circumvent the problem of having only subsidized new homeowners from the IFL files, we use exhaustive tax files about French homeownership (*Fichiers Fonci-ers*) to determine the total number of new homeowners, independently of whether they are subsidized. This variable is much more informative for extensive margin effects, as increasing the number of recipients without affecting the number of homeowners would correspond to opportunistic behaviour. Using the temporal dimension of these administrative data, we identify first-time homeowners as defined by the IFL policy. Hence, we can disentangle an increase in the number of homeowners from an increase restricted to recipients only. We obtain for each municipality the number of first-time owners, who were eligible to the IFL, by counting the number of homeowners that were absent from the tax file in the previous two years.¹³ Although the tax files and IFL files are independent databases, the total number of first-time homeowners estimated from tax files is always higher than the number of contracts from IFL files (except for two municipalities that are removed from our sample). Finally, we recover the total number of newly-built housing over the 2010–2013 period based on the construction year reported in the tax files.

Transaction data. We use a third exhaustive individual dataset on housing transactions (DV3F) to compute, at the municipal level, the average unitary price of housing over the pre-treatment period 2010–2013. To mitigate border effects in the delineation of housing markets and tightness, we also compute, for each municipality, the averages of the unitary prices over the same pre-treatment period for neighbouring municipalities using the spatial contiguity definition. These data also allow building variables related to the post-treatment outcomes for the overall housing markets, including average housing prices, average surfaces, and average unitary prices for the three periods.

Socio-demographic data. For each municipality under consideration, we obtain from the *French National Institute of Statistics and Economic Studies* (INSEE) (year 2013) data before the ABC reform in 2014 on population density, median income, and socio-professional categories.

¹³The policy defines first-time owners as individuals being renters of their primary residence for at least two years purchasing a housing unit for residence.

Sample. Our sample is composed of all French municipalities. We assign each municipality to its ABC level and report descriptive statistics on the intensity of IFL treatment, mortgage conditions, and pre-treatment variables in [Table 1](#).

The IFL treatment variables are ordered according to the ABC zoning. Indeed, the A-tier benefits from the highest maximum of IFL, whereas the C-tier benefits from the lowest regardless of the period of interest. Moreover, according to a set of pre-treatment variables (Panel A), the A-tier exhibits the highest housing price, unitary housing price, inhabitants density or number of new housing units whereas the C-tier from the lowest. Hence, the treatment intensity is not endogenously defined.

Table 1. Main pre-treatment and treatment variables for municipalities along the ABC zoning

Variable	Period	Country	ABC Zoning Areas			
			A	B ₁	B ₂	C
A - Pre-treatment variables						
Number of Municipalities (thousand of units)	2013	34.970	0.724	1.535	3.828	28.883
		100%	2.07%	4.39%	10.95%	82.59%
Housing Price (thousand euros)	2010–2013	153.1 (68.0)	284.0 (124.8)	234.9 (74.8)	188.6 (59.6)	139.0 (54.9)
Unit. Housing Price (euros by squared meter)	2010–2013	1,608.7 (691.8)	3,558.7 (1,054.5)	2,597.0 (557.5)	2,003.5 (569.3)	1,430.9 (502.6)
Unit. Price of Neighbors (euro per squared meter)	2010–2013	1,561.0 (724.0)	3,654.1 (1,099.1)	2,618.6 (587.2)	1,975.2 (582.8)	1,371.6 (517.1)
Population Density (inhabitants by hectare)	2013	1.9 (8.1)	26.4 (38.2)	6.8 (9.4)	3.1 (4.6)	0.7 (1.0)
Median Household Income (thousand euros by year)	2013	20.0 (3.4)	25.3 (6.0)	24.0 (4.8)	22.1 (3.7)	19.2 (2.6)
Number of New Housings (number of units)	2010–2013	41.9 (226.7)	364.5 (918.7)	226.8 (648.3)	72.4 (151.7)	15.9 (28.7)
B - IFL Policy Values						
Maximum Ceiling Value (thousand of euros)	2015	-	150.0	135.0	110.0	100.0
	2016–2017	-	150.0	135.0	110.0	100.0
	2018–2019	-	150.0	135.0	110.0	100.0
Maximum Coverage Share (percent)	2015	-	26.0	26.0	21.0	18.0
	2016–2017	-	40.0	40.0	40.0	40.0
	2018–2019	-	40.0	40.0	20.0	20.0
Maximum IFL Amount (thousand of euros)	2015	-	39.0	35.1	23.1	18.0
	2016–2017	-	60.0	54.0	44.0	40.0
	2018–2019	-	60.0	54.0	22.0	20.0
Average Subsidy (thousand euros)	2015	5.21	10.31	9.41	5.81	4.02
	2016–2017	9.63	13.96	12.35	10.45	8.99
	2018–2019	5.03	12.44	10.89	4.64	3.98
C - Mortgage Market Conditions						
Mortgage Maturity (percent)	2015	228	244	254	238	221
	2016–2017	260	268	269	265	258
	2018–2019	258	267	267	262	255
Annual Interest Rate (number of months)	2015	2.51	2.46	2.45	2.52	2.52
	2016–2017	1.87	1.77	1.77	1.89	1.88
	2018–2019	1.62	1.53	1.52	1.65	1.63

Notes: French municipalities are classified according to the ABC zoning in columns. Panel A reports the average and standard deviation of pre-treatment variables used to control the endogenous treatment assignment. The first three variables of panel B correspond to the IFL parameters for each period (constant between municipalities) with a Maximum IFL Amount that equals the maximum ceiling value times the Maximum Covering Share. The Average Subsidy is computed from IFL data and Equation 1. Panel C reports the average of loan maturities and interest rates, also extracted from IFL files.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

3 Empirical Framework

We consider the objective of the IFL policy from a policymaker's perspective in terms of extensive and intensive margins. Policymaker objectives are proxied through the marginal valuation of, respectively, one additional homeowner and recipient purchase value. We detail our extensive and intensive margin definitions related to public spending according to a return on investment function. We first present an overall treatment effect and link it to policy treatment variation (i.e. ceiling value, covering share and mortgage conditions). Finally, we highlight the needed statistics to compute the return on investment concerning the maximization of specified policy objectives.

3.1 Outcome variables

We develop a framework for evaluating IFL policy that is derived from the objectives set by policymakers. On the one hand, the total number of new homeowners (denoted N) is likely to be the most relevant variable according to the policy design. We also consider the number of IFL recipients $\tilde{N}$, although it does not correspond to the full extensive margin, as increasing $\tilde{N}$ without changing N cannot be considered as favouring homeownership that we define as an opportunistic behaviour. On the other hand, the intensive margin captures the distortion of recipients' housing choices. The recipients could use the subsidy to increase their purchase price (noted $\tilde{V}$), either by increasing the area (noted $\tilde{S}$) or the price per area (noted $\tilde{P}$). Finally, policy-induced demand could affect the average housing price from capitalization effects, especially in the context of low supply elasticity. Hence, the increase in housing demand (that could be proxied by the number of recipients $\tilde{N}$) might affect the entire local population through the average housing price V . We then consider that the potential capitalization effect could in return, affect both extensive and intensive margin effects.

Beyond the extensive and intensive margin effects, variation in the IFL intensity will affect the cost for the policymaker. The policy cost for the administration depends on the number of recipients (denoted $\tilde{N}$) and the average cost per recipient (denoted $\tilde{C}$). Hence, the effect of treatment intensity variation on the cost for the administration is not clear-cut as it depends both on first-time owners' behaviour (whether they benefit from the policy) in addition to the treatment intensity per recipient. Finally, we assume that policymakers aim to achieve their policy objectives (support homeownership development) while minimizing the overall cost of the policy.

3.2 Policy objectives

We consider that policymakers' objectives can be summarized in an efficiency measure using the monetary valuation of both extensive and intensive margin effects. We then define the policy return on investment, denoted E , that derives from policymakers' objective as follows

$$E = \phi N + \psi \tilde{V} - \tilde{N} \tilde{C} \quad (2)$$

with ϕ and ψ representing respectively the policymaker's valuation of an additional homeowner and the distortion of recipients' housing choice. E represents the monetary valuation of the IFL policy effects, taking into account the total costs defined by the product of the number of recipients ($\tilde{N}$) and the average cost per recipient ($\tilde{C}$). We consider the case where policymakers seek to invest through an increase in policy variables a , i.e. covering share, ceiling value and mortgage conditions. We derivate Equation 2 according to the variation in policy variable a that defines treatment intensity (covering share, ceiling value, and mortgage conditions) such as $e_a = \partial E / \partial a$

$$e_a = \frac{\partial E}{\partial a} = \phi \left(\frac{\partial N}{\partial a} + \frac{\partial N}{\partial V} \cdot \frac{\partial V}{\partial a} \right) + \psi \left(\frac{\partial \tilde{V}}{\partial a} + \frac{\partial \tilde{V}}{\partial V} \cdot \frac{\partial V}{\partial a} \right) - \left(\tilde{N} \frac{d\tilde{C}}{da} + \tilde{C} \frac{d\tilde{N}}{da} \right) \quad (3)$$

The sign of e_a is the main criterion to determine whether increasing public spending by treatment variable a is positive according to the policy objectives. Indeed, if e_a is positive, the policymaker would consider the benefits to be greater than the costs and public expenditure, leading to a positive return on investment. The return on investment function is composed of a direct and an indirect effect at both extensive and intensive margins with $N = N[a, V(a)]$ and $\tilde{V} = \tilde{V}[a, V(a)]$. On the one hand, either the number of homeowners or the recipients' housing choices might vary according to treatment intensity variation (the direct effect). On the other hand, potential price capitalization resulting from treatment variation might in return reduce either the likelihood of transitioning to homeownership or affect recipients' housing choices (the indirect effect).¹⁴ Hence, the return on investment for additional public spending depends at both margins on the direct (treatment response, denoted $\frac{\partial}{\partial a}$) and the indirect (adaptation to potential price capitalization, denoted $\frac{\partial}{\partial V} \cdot \frac{\partial V}{\partial a}$) effects. Finally, we consider that the policy cost only depends on variation in average cost per recipient and the number of recipients, which motivates this simple framework in terms of derivatives.

We distinguish two sets of values from Equation 3. On the one hand, it requires observing the implicit monetary values set by policymakers to reflect their policy objectives. Al-

¹⁴For instance, see Waxman et al. (2020) for housing choices responses to housing affordability issues.

though some papers address the valuation of externalities associated with homeownership (Coulson and Li, 2013), it is not clear how policymakers value intensive margin effects relative to extensive margin ones. We then discuss the return on investment of increasing public spending through IFL subsidies, following scenarios that differ according to the valuation of extensive and intensive margin effects. In reasonable scenarios, an additional homeowner is positively valued as it is the policy’s stated objective. The valuation of modification of recipients’ housing choices is less straightforward, leading us to introduce more variation in the credible valuation range. On the other hand, first-order responses to treatment variable variation for main outcomes such as the number of homeowners, housing market price or recipients’ purchase price are needed. It composes our building blocks to estimate the return on investment of raising IFL subsidy through each treatment variable. Moreover, we must estimate a similar quantity for average spending per recipient and the number of recipients to assess the impact on policy cost. To estimate these first-order responses, we leverage differences in treatment intensity following the ABC zoning to obtain dose-response functions.

3.3 Dose-response functions

We recover the marginal effects at both extensive and intensive margins for the IFL policy from the counterfactual framework (Rubin, 1974), through dose-response functions relating policy-relevant treatment effects to the four policy variables. Variations in IFL policy across the four ABC zones and the three periods define a multivalued treatment taking $G = 12$ levels. Let $g = 1 \dots G$ denote a level of treatment and T_g a dummy variable that indicates whether the municipality receives this level. Then, we have:

$$Y = \sum_{g=1}^G T_g Y_g, \quad (4)$$

where Y is the observed outcome, equals to its potential value Y_g only if a municipality receives treatment g . The main outcomes of interest are $Y = N$ for the extensive margin, $Y = V$ for the intensive margin, and $Y = \tilde{C}$ for budget costs, while $Y = \tilde{V}$ informs about unintended effects on consumption. Each bilateral combination of different treatment levels g and g' corresponds to a variation of at least one policy treatment variable. We exploit this structure of the IFL policy to map policy-relevant treatment effects to treatment variable variations. Considering the requirement of first-order derivatives to estimate Equation 3, we retain a set of linear dose-response functions for each outcome Y with:

$$\mathbb{E}(Y_g - Y_{g'}) = \beta_0^Y + \sum_a \beta_a^Y (a_g - a_{g'}) + \xi. \quad (5)$$

with mean-independent errors ξ . The Ordinary Least Squares (OLS) coefficients β_a^Y provide a summary of the effects of treatment variables a on the heterogeneity of treatment effects $\mathbb{E}(Y_g - Y_{g'})$ and allow recovering the derivatives of efficiency measures (with $\beta_a^Y \equiv \partial Y / \partial a$ in Equation 3). For outcomes concerning the whole population, $Y \in \{N, V, P, S\}$, the average treatment effects (ATEs) are policy relevant as they appear in the left-hand side of Equation 5. Indeed, ATEs represent the change of Y caused by the policy g relatively to g' for the whole population and β_a^Y summarizes how these changes can be attributed to differences between a_g and $a_{g'}$ (the policy variables and mortgage conditions) that we observe. For outcomes affecting only recipients, $Y \in \{\tilde{N}, \tilde{V}, \tilde{P}, \tilde{S}, \tilde{C}\}$, the policy-relevant treatment effects concern recipients (ATT). The left-hand side of the dose-response function (Equation 5) is then $\mathbb{E}(Y_g - Y_{g'} \mid T = g)$. As we study bilateral combinations *within* three distinct periods, this gives $4 \times (4 - 1) \times 3 = 36$ policy-relevant treatment effects. Therefore, each set of dose-response functions is estimated based on 36 observations for each of the nine outcomes. This allows us to recover the marginal effect at both margins and estimate the return on investment for additional public spending. Yet, to estimate dose-response functions, we need to first compute the left-hand side of the Equation 5, which represents outcome counterfactual differences when switching treatment intensity from one level to another (e.g. A to B₁).

4 Identification strategy

4.1 Identifying Assumptions

Facing the endogeneity of the ABC zoning due to the definition of housing market tightness, we maintain two assumptions to recover causal treatment effects. The first is that, conditionally on pre-treatment variables, the g treatments are weakly unconfounded.

Assumption 1 *Weak Unconfoundedness.*

$$\forall (g, \mathbf{X}) Y_g \perp T \mid \mathbf{X}$$

According to this assumption, the set of pre-treatment variables $\mathbf{X}$ ensures conditional randomization of the IFL policy between municipalities. This selection-on-observables restriction considers that all the structural differences between municipalities are controlled by pre-treatment variables and that the differences between the conditional outcomes can only be attributed to policy changes. As g describes both spatial and time variations, we use this assumption both between areas of ABC zoning and between policy periods.

The well-known property of dimension reduction of well-specified propensity scores (Hahn,

1998) allows to parsimoniously model the conditional expectation of the outcomes, as long as we have $Y_g \perp T | p_g(\mathbf{X})$ with $p_g(\mathbf{X}) \equiv \mathbb{P}(T = g | \mathbf{X})$ from Assumption 1. This is the definition of the Generalized Propensity Score (GPS, Imbens, 2000) as the probability of receiving a level of treatment knowing the pre-treatment variables. As Crump et al. (2009) show, the propensity to receive treatment should not be too close to zero or one to ensure precise and robust estimates. This leads to the following overlap assumption:

Assumption 2 *Overlap*

$$\forall(g, \mathbf{X}), p_g(\mathbf{X}) > 0$$

Under the two previous assumptions, Słoczyński and Wooldridge (2018, Lemma 3.2) show that counterfactual treatment effects can be identified from the data at hand. The average outcome $Y_{g'}$ for a counter-factual treatment level g' , respectively for the whole population and for municipalities that receive the treatment level g , are respectively:

$$\mathbb{E}(Y_{g'}) = \mathbb{E} \left[\frac{T_{g'}}{p_{g'}(\mathbf{X})} Y \right] \quad \text{and} \quad \mathbb{E}(Y_{g'} | T = g) = \frac{1}{\mathbb{P}(T = g)} \cdot \mathbb{E} \left[\frac{p_g(\mathbf{X})}{p_{g'}(\mathbf{X})} T_{g'} Y \right]. \quad (6)$$

These statistics concern, respectively, the full population of homeowners impacted by the externalities at both margins and the recipients targeted by the policy support. They are the building blocks of the policy-relevant treatment effects under consideration, as the ATE of g instead of g' on the outcome Y is $\mathbb{E}(Y_g) - \mathbb{E}(Y_{g'})$ and the related ATT is $\mathbb{E}(Y_g | T = g) - \mathbb{E}(Y_{g'} | T = g)$. These counterfactual statistics are used to build policy-relevant treatment effects as they are related to different populations.

4.2 Specification of the Propensity Score

The propensity score allows estimating $p_g(\mathbf{X})$ to recover treatment effects from Equation 6. In line with the notion of housing market tightness, we define an unobserved latent variable η_i^* crossing thresholds to model the classification of municipalities. The propensity for a municipality i to be high in the hierarchy depends on the J pre-treatment variables x_{ji} used to proxy the political decision, a bivariate smooth function of the geographical coordinates of its centroid $\mathbf{z}_i$ (longitude and latitude, Gilbert et al., 2024), and a random term ε_i representing the arbitrary part of the zoning explained in the Section 2.2. This latter term is assumed to be logistically distributed to produce an ordered logit model. The latent variable describing the tightness of the housing market η_i^* is then:

$$\eta_i^* = \alpha + \sum_{j=1}^J f_j(x_{ji}) + h(\mathbf{z}_i) + \varepsilon_i. \quad (7)$$

The J univariate functions f_j are specified as additive spline transformations of pre-treatment variables, which corresponds to the generalized additive model framework (GAM, Wood, 2017). The spline coefficients are shrunk endogenously by penalized iterated weighted least squares while the smoothing parameters are estimated using a separate criterion from the restricted maximum likelihood (Wood, Pya and Säfken, 2016). The same estimation procedure is used simultaneously for the bivariate smooth function h of coordinates, the main difference is the *a priori* specification of the spline that is a bivariate thin plate. Whereas the geographic regression discontinuity restricts the sample to observations close to the boundary to assume that the treatment is quasi-random between treatment and control groups, our approach explicitly models the spatial dimension in the treatment assignment and recovers the arbitrary part of the treatment assignment, making the best of the full sample.

By noting Λ the cumulative function of the logistic distribution and $\mu_0 < \mu_1 < \dots < \mu_5$ the unknown ordered thresholds related to the four ABC zones, the GPS for the IFL policy are (with $\eta_i \equiv \eta_i^* - \varepsilon_i$ the deterministic parts of the latent variable):

$$p_g(\eta_i) = \Lambda(\mu_g - \eta_i) - \Lambda(\mu_{g-1} - \eta_i). \quad (8)$$

Because municipalities designed as A are tighter than others (B_1, B_2, C) and because η_i is a measure of market tightness, values of the latent variable lie between the thresholds μ_4 and μ_5 . As the ABC zoning did not change in the 2015–2019 period, the probability of being in a given zone is constant over time. Then, an appealing property of the ordered structure of the ABC zoning is that, if the GPS is well specified, conditioning on the deterministic part of the latent variable η_i is sufficient to reach weak unconfoundedness (instead of the full set of pre-treatment variables $\mathbf{X}$). The potential remaining threats for GPS specification concern omitted variable bias. Despite the spatial smoothing functions, it might not fully capture unobservable variables with variations occurring at the finest spatial level (i.e. from one street to another), i.e. air pollution due to road traffic or school quality. We consider the likelihood of such omitted variables and their impact on the GPS specification to be very marginal. However, to prevent GPS misspecification, we favour a doubly robust estimation relying on a specification of the outcomes. In this case, the estimated coefficients are consistent if at least one specification is well specified (Robins and Rotnitzky, 1995; Słoczyński and Wooldridge, 2018).

4.3 Specification of the Outcomes

The outcomes are specified using the same semi-parametric GAM framework. Recall that we introduce, as control variables in the outcome specification pre-treatment, variables used in the GPS specification to provide a doubly robust estimator, which reinforces

the credibility of our results. The main difference is that each outcome Y is modelled separately for each subsample defined from the treatments g received by the municipalities. The smooth functions f_j and h are now indexed by the outcome y and the treatment g such that:

$$y_{gi} = \alpha_g^y + \sum_{j=1}^J f_{gj}^y(x_{ji}) + h_g^y(\mathbf{z}_i) + \varepsilon_{gi}^y. \quad (9)$$

The same pre-treatment variables and geographical coordinates are used, with different smoothing parameters. As we have nine outcomes, four treatment levels and three periods, Equation 7 corresponds to 108 GAM estimations to obtain the full set of functions f_{gj}^y and h_g^y for a given GPS. Following Słoczyński and Wooldridge (2018), the double robustness property requires that outcome regressions are weighted according to GPS ratios as in Equation 6. To recover the average counter-factual outcome for the treatment g' for the municipalities receiving g , each municipality is weighted by $\hat{p}_g(\eta_i)/\hat{p}_{g'}(\eta_i)$ predicted from the first stage. As generally advised in the literature, we use normalized weights by dividing them by their sum within each treatment subsample.

We close this section with the formulas that we use to assess the return on investment of the IFL policy. The counter-factual building blocks of Equation 6 are recovered from the regression of the outcome Y on the subsample of municipalities with treatment g' using respectively $1/p_{g'}(\eta_i)$ and $p_g(\eta_i)/p_{g'}(\eta_i)$ as weights. Under assumptions 1 and 2, noting $\mu_g \equiv \mathbb{P}(T = g)$ the share of municipalities that receive the treatment g , averaging the fitted values provides a consistent estimation as:

$$\mathbb{E}(Y_{g'}) = N^{-1} \times \sum_{\ell=1}^N \hat{y}_\ell(g') \quad \text{and} \quad \mathbb{E}(Y_{g'} | T = g) = \mu_g^{-1} \times \sum_{\ell=1}^N T_{g\ell} \times \hat{y}_\ell(g') \quad (10)$$

where $\hat{y}_\ell(g') \equiv \hat{\alpha}_{g'}^Y + \sum_{j=1}^J \hat{f}_{g'j}^Y(x_{j\ell}) + \hat{h}_{g'}^Y(z_\ell)$ comes from the estimation of the outcome Y for the subset of municipalities that receive treatment g' . The predicted outcome values for the entire population of municipalities with $\ell = 1, \dots, N$, will be used as the outcomes in Equation 5.

5 Results

We first present the estimation results for the GPS of each municipality, followed by the estimation of policy-relevant treatment effects and dose-response functions. We close this section with our measures of return on investments resulting from IFL subsidy variations according to each policy variable, i.e. variables that affect treatment intensity such as the

ceiling value, the covering share and the mortgage conditions.

5.1 GPS Estimation

We include the geographical coordinates of the centroids of each municipality using bivariate smoothing splines to control for spatial confounders (Gilbert et al., 2024). As allowed by the GAM framework, all variables enter semi-parametrically with a degree of smoothing that is endogenously shrunk by the penalized estimation procedure. Table 2 provides the joint significance of the spline transformations of each variable according to different specifications and maximum degree of spatial smoothing, while detailed results on GPS estimation are reported from Appendix C.1 to Appendix C.2.

Table 2. Covariates' joint significance from first-stage ordered GAMs

Max. degrees of freedom	<i>Outcome: Ordered ABC Zoning</i>					
	No Spatial Smoothing		With Spatial Smoothing			
	df = 0	df = 0	df = 50	df = 50	df = 100	df = 200
Population Density	1,991.3*** [6.1]	1,723.1*** [5.7]	2,003.3*** [5.8]	1,656.4*** [5.8]	1,688.5*** [6.0]	1,479.1*** [6.0]
New Housing Unit	468.7*** [6.0]	99.0*** [5.3]	295.2*** [5.3]	126.1*** [5.0]	127.4*** [4.9]	141.2*** [4.8]
Median Annual Income	1,647.6*** [6.7]	353.5*** [6.6]	654.7*** [6.7]	208.4*** [6.2]	200.5*** [6.1]	182.7*** [6.0]
Professional Occupations	984.1*** [37.0]	819.4*** [28.4]	312.9*** [30.6]	317.3*** [32.6]	273.8*** [25.1]	267.8*** [26.8]
Unitary Housing Price		214.8*** [6.6]		70.3*** [5.5]	67.7*** [5.2]	51.3*** [1.0]
Neighboring Unitary Price		110.4*** [1.1]		37.1*** [3.1]	26.6*** [1.0]	23.1*** [4.2]
Spatial Coordinates			4,018.4*** [47.9]	2,211.0*** [47.3]	2,575.9*** [90.2]	3,048.8*** [165.2]
Number of Observations	26,818	26,818	26,818	26,818	26,818	26,818
McFadden R2	52.60	61.31	67.08	69.17	70.98	73.81
Percent of Good Predictions	85.88	87.31	88.94	89.29	89.70	90.13
Akaike Information Criterion	18,625.4	16,406.0	14,602.8	14,152.3	13,736.9	13,156.0

Notes: The top panel reports χ^2 statistics of joint significance for each covariate of the first-stage GPS. Professional Occupations are coded as population shares of eight categories according to the one-digit French *Catégories Socio-Professionnelles*. The effective degrees of freedom reported in brackets indicate the smoothing intensity, low values correspond to more smoothing. The unit of observation is the French municipality, columns report different specifications with different covariates and different maximum spatial smoothing. Estimations come from the `gam` function of the `mgcv` R package (Wood, Pya and Säcken, 2016).

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

*** $p < 0.01$, ** $p < 0.05$ * $p < 0.1$.

The specification with the lowest spatial smoothing (that allows the consideration of spatial heterogeneity on a fine scale, reported in the last column) yields 90.1% of correct predictions of the ABC classification for municipalities. This indicates the relevance of

the ordered framework for modelling the ABC classification and increases the likelihood of having a well-specified GPS. Although models with higher maximum degrees of freedom allowed for spatial coordinates to obtain even better predictions (91.2% for $df = 400$), their computational cost and the issues of dimensionality for our smallest sample lead us to prefer more parsimonious specifications. Nevertheless, our main results regarding the effects of IFL policies are robust to the specification of the maximum degrees of freedom for spatial coordinates, albeit increasing the maximum degree of freedom for spatial smoothing significantly reduces standard errors. In particular, the introduction of spatial coordinates affects the joint significance associated with the unitary housing price being consistent with the local characteristics of the housing market. The joint significance is the highest of the pre-treatment variables, confirming our expectation about the presence of unobservable spatial variables. In our preferred specification ($df = 200$), the contribution of the unitary housing price is linear and increasing, consistent with the ABC perimeter definition.

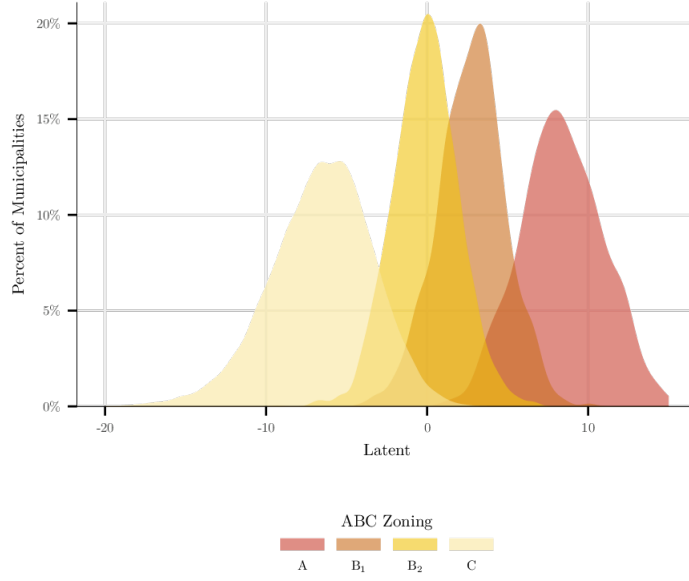
Although prediction errors are limited, they concern municipalities commonly used for geographic regression discontinuity design, as 77.2% of errors have at least one neighbouring municipality being classified in another treatment level. In line with the regression discontinuity assumptions, for these municipalities, the treatment assignment is quasi-random, which increases the likelihood of wrong predictions for these municipalities. In addition, our prediction errors mainly concern municipalities that changed ABC-tier during the 2014 reform. Similarly, we might expect the treatment assignment to be less predictable.

Since overlap is crucial to recover consistent effects but more difficult to achieve for high-dimension model variables (D’Amour et al., 2021), we compare the distribution of the latent variable underlying the classification process (Fig. 2). Latent distributions follow the ordered structure of the ABC classification, as consecutive treatment levels have greater common support than non-consecutive ones. Although the overlap is reduced, there is still common support for extreme levels. This is probably due to the spatial proximity of some A- and C-tier municipalities. However, although treatment assignment is based on the characteristics of the municipality, it still contains some arbitrariness, which we exploit for identification.

5.2 Overall Policy Effects

We now assess the relevance of our control variables in the outcome specification using pooled models for our doubly robust estimator. Considering the large set of pre-treatment variables and our nine outcomes, we only report the joint significance of each pre-treatment variable in Table 3 to assess the statistical power of pre-treatment variables.

Figure 2. Overlap between predictions of tightness between the different ABC zones



Notes: The distributions of the latent tightness variable (x-axis) are predicted from the first stage GPS with a maximum degree of freedom set to 200 (6th column of Table 2). As a latent variable, $\hat{\eta}^*$ is unitless and is displayed between municipalities according to the ABC classification. We report the distribution *within* each classification level (rather than the distribution of the all population) for clarity reasons.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

Pre-treatment variables introduced as regression adjustment in the pooled models explain more than 74% of the observed variance in the number of first-time owners. The development of housing supply, measured by the number of new housing, is highly significant in explaining both the number of first-time owners and the number of recipients. In addition, the local housing market price and median income are significant in explaining the number of transitions to homeownership, highlighting the importance of affordable housing for transitions to homeownership. It supports our approach of introducing pre-treatment variables in the regression adjustment to control for potential heterogeneity.

5.3 Treatment Effects and Dose-Response Functions

From the second step, we estimate combinations of bilateral effects (g, g') to infer required statistics for the estimation of efficiency first-order response to treatment variable variations (Equation 3). We report these bilateral estimations and standard errors, using a bootstrap approach with 500 iterations, respectively, in Appendix C.4 for ATE and Appendix C.5 for ATT. They constitute our building blocks to estimate dose-response functions according to each treatment variable for both extensive and intensive margin outcomes.

Table 3. Covariates' joint significance from second-stage pooled GAMs

	Outcome variables from...								
	Tax	Transaction Data				IFL Files			
	N	V	S	P	$\tilde{N}$	$\tilde{V}$	$\tilde{S}$	$\tilde{P}$	$\tilde{C}$
Population Density	517.1*** [8.7]	9.6*** [6.6]	102.7*** [8.6]	50.8*** [7.8]	51.9*** [8.7]	17.6*** [6.8]	88.0*** [7.7]	58.9*** [7.9]	32.2*** [4.2]
Number of New Housing	2,380*** [8.2]	38.7*** [5.6]	93.3*** [4.6]	120.1*** [5.6]	1,128*** [7.1]	8.8*** [4.3]	11.5*** [4.3]	10.6*** [4.1]	3.0** [4.1]
Median Income	53.4*** [8.4]	68.4*** [7.0]	198.2*** [6.8]	3.3** [3.0]	16.9*** [6.0]	105.3*** [6.5]	27.3*** [5.7]	22.3*** [8.5]	1.5** [1.8]
Professional Occupations	771.3*** [50.8]	55.8** [34.1]	21.3** [49.2]	48.6** [38.7]	357*** [41.4]	6.6** [38.3]	6.6** [32.9]	12.1** [47.0]	34.0** [20.4]
Lagged Unitary Price	9.7*** [8.2]	21.7*** [4.5]	8.3*** [8.2]	31.9*** [3.5]	9.7*** [9.0]	4.7*** [3.1]	2.8*** [6.4]	3.8*** [8.6]	4.5*** [7.6]
Lag. Neighbor. Unit. Price	6.7*** [8.3]	96.7*** [7.5]	11.7*** [7.2]	83.7*** [8.1]	9.1*** [8.8]	10.1*** [8.7]	18.3*** [7.3]	32.8*** [7.6]	7.1*** [3.6]
Spatial Coordinates	33.7*** [189]	16.6*** [182]	37.6*** [188]	8.3*** [179]	22.0*** [186]	21.8*** [181]	8.5*** [168]	20.5*** [193]	7.0*** [112]
Number of observations	54,993	54,993	54,993	54,993	54,993	54,993	54,993	54,993	54,991
McFadden R2	77.72	56.12	36.53	56.99	54.52	45.27	18.63	55.81	9.16

Notes: For the nine outcomes of interest (in columns), the table reports the F statistics for the joint significance of each covariate (in rows). N accounts for the number of new homeowners, V for housing value, S for surface, and P for unitary housing price. The variables with a $\tilde{\cdot}$ are the same variables computed for IFL recipients, and $\tilde{C}$ is the IFL cost. We report pooled GAMs for all treatment levels for the sake of clarity, different GAMs are estimated for each treatment level in the policy-relevant treatment effects reported in the text. Professional Occupations are coded as population shares of eight categories according to the one-digit French *Catégories Socio-Professionnelles*. The effective degrees of freedom reported in brackets indicate the smoothing intensity, low values correspond to more smoothing. The unit of observation is the French municipality and the maximum degree of freedom we allow for the spatial coordinates is 200.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

*** $p < 0.01$, ** $p < 0.05$ * $p < 0.1$

We estimate each β_a^Y from Equation 5 by regressing the bilateral combinations of the treatment level on the differences in treatment variable values between treatment level g and g' . Since we consider small treatment variable variations, our dose-response functional form is linear. In addition, our weighting scheme to estimate the relevant dose-response parameters depends on the nature of the estimands. We weigh bilateral combinations according to the number of municipalities receiving treatment level g for ATT estimands. We do not introduce weights for the ATE as it concerns the entire population, unlike ATT. Results from linear dose-response specifications, estimated by WLS (for ATT) and OLS (for ATE), are reported in Table 4. Standard errors are estimated using bootstrap with 500 iterations.

Despite a significant effect on policy costs for the covering share (column 9, Table 4), which is the main treatment variable used for policy reforms, it has no significant effect on the number of homeowners (column 1, Table 4). Increasing the covering share is unlikely to achieve the main policy objective, as the number of homeowners does not increase significantly with this variable. Meanwhile, increasing the ceiling value (the second

Table 4. OLS coefficients for policy treatment variables from dose-response functions

	Outcome variables from ...								
	Tax (ATE)	Transaction Data (ATE)			IFL files (ATT)				
	N	V	S	P	$\tilde{N}$	$\tilde{V}$	$\tilde{S}$	$\tilde{P}$	$\tilde{C}$
Covering Share	-0.007 (0.025)	0.032** (0.015)	0.004 (0.006)	0.007 (0.015)	-0.033 (0.046)	0.001 (0.007)	0.010 (0.011)	-0.000 (0.011)	0.041*** (0.011)
Ceiling Value	-0.024 (0.026)	-0.003 (0.015)	-0.006 (0.005)	-0.004 (0.014)	0.091** (0.035)	-0.013 (0.009)	-0.047*** (0.010)	0.026*** (0.008)	-0.006 (0.010)
Interest Rate	-0.004 (0.007)	0.001 (0.004)	-0.001 (0.001)	-0.003 (0.004)	0.021** (0.010)	-0.005*** (0.002)	-0.009*** (0.003)	0.004 (0.002)	-0.001 (0.003)
Loan Maturity	0.009 (0.012)	-0.002 (0.011)	-0.001 (0.004)	-0.004 (0.009)	-0.025 (0.021)	0.004 (0.003)	0.025*** (0.005)	-0.016*** (0.004)	0.021*** (0.006)
Price	0.003 (0.005)					-0.001 (0.003)			
Constant	-0.038 (0.052)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.130 (0.077)	0.009 (0.021)	-0.039 (0.023)	0.035* (0.020)	-0.014 (0.023)
R ²	0.162	0.569	0.505	0.649	0.522	0.334	0.456	0.434	0.906
Adj. R ²	0.022	0.513	0.441	0.603	0.460	0.223	0.386	0.361	0.894
N	36	36	36	36	36	36	36	36	36

Notes: For the nine outcomes Y in columns, the table reports the β_a^Y coefficients associated to each treatment variable in rows. They are estimated from dose-response functions of Equation 5. N accounts for the number of new homeowners, V for housing value, S for surface, and P for unitary housing price. The variables with a $\tilde{\cdot}$ are the same variables computed for IFL recipients, and $\tilde{C}$ is the IFL cost per recipient. The interest rate is expressed in a hundredth per cent. The unit of observation is the bilateral combination of four ABC zones for the three periods of interest, the full set of policy-relevant treatment effects is reported in the Appendix C.4 and Appendix C.5 of OA. Standards errors in parentheses are estimated using bootstrap with 500 iterations accounting for the uncertainty of treatment effects. ATEs for tax and transaction data are weighted according to the inverse of their bootstrapped standard errors, and ATT for IFL variables are additionally weighted according to the number of municipalities receiving the considered treatment levels.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

*** $p < 0.01$, ** $p < 0.05$ * $p < 0.1$

treatment variable that policymakers can control) has no significant effect on the number of homeowners, while it increases the number of policy recipients (+9.1%). Given the joint effect of the ceiling on the number of homeowners and recipients, raising the ceiling value is more likely to induce opportunistic behaviour than to cause homeownership. In addition, it should be noted that the impact of IFL on tenure decisions is also likely to be independent of the characteristics of the credit market, while opportunistic behaviour is more likely to be favoured in a context where credit conditions are less favourable for households (higher interest rates).

Whereas the covering share has at best a weak effect on the number of homeowners, it increases housing prices for the entire local housing market. A one-unit increase in the covering share raises the price of all transactions by 3.2%, leading to a significant unintended effect on the housing market. The variation in subsidy resulting from the covering share might increase housing demand whereas it does not distort recipients' housing choices. In contrast, raising the ceiling value has no significant effect on the local

housing price, while the effect on recipients' housing choices is more contrasted. Indeed, while it negatively impacts housing size (columns 6 and 7, [Table 4](#)), the price per area unit covaries positively with the ceiling value (column 8). As raising the ceiling value increases treatment intensity only for the more expensive operations (for more details, see [Appendix B](#)), the distortion of housing choices might correspond either to an increase in location quality, construction quality or both. However, as the policy affects the number of recipients, we cannot exclude the risk that sample selection biases our results at the intensive margin. Assuming that switchers are likely to be more financially constrained, we expect our results at the intensive margin to be a lower bound.

Similarly, recipients' housing choices are sensitive to mortgage conditions. On the one hand, recipients decrease housing size when interest rates increase. We assume that the increase in treatment intensity due to interest rates is outweighed by the additional interest they must pay (recall that the IFL is a complementary loan that cannot overcome 40% of the overall borrowed amount). On the other hand, longer loan maturity increases the housing unit recipients purchase, whereas it decreases the average price per square meter decrease.

These results are consistent with alternative results obtained by exploiting a natural experiment. Precisely, using difference-in-differences (see [Appendix D](#)), we focus on the 2018 reform, which changes the coverage share for two ABC tiers while leaving this policy unchanged for two others. The findings imply that reducing the coverage share has a significant effect on the cost of the policy but no significant effect on the number of homeowners.

5.4 Simulation According to Policy Objectives

We finally provide counterfactual simulations that define return on investment for increasing public spending according to each treatment variable. For comparison purposes, we choose to simulate treatment variable variations (ceiling value, covering share and mortgage conditions) that have a similar impact on the total cost of the policy and provide normalized measures for one additional euro of public spending. The statistics needed to estimate [Equation 3](#) are obtained from the dose-response functions ([Table 4](#)). Note that the effect of the treatment variable variation a on the total cost of the policy aggregates the effect on the cost per recipient ($\partial\tilde{C}/\partial a$) and the effect on the number of recipients ($\partial\tilde{N}/\partial a$). Then, despite a non-significant effect on the cost per recipient of raising the ceiling value ([Table 4](#)), it has a positive effect on the total cost of the policy, given the strong effect on the number of recipients. It should be noted that the effect on total cost is positive regardless of the treatment variable, although it is not statistically significant for some treatment variables.

As we do not observe the marginal valuation of both extensive and intensive margin effects from a policymaker perspective, we perform various simulations to assess under which conditions increasing the subsidy provides a positive return on investment. Yet, we only consider cases where increasing the number of homeowners is positively valued, as this is the main objective to generate additional homeowners. For the intensive margin valuation, the valuation is not straightforward from a policymaker's perspective. On the one hand, policymakers might aim to support recipients' utility by enhancing housing size. On the other hand, considering raising issues about land consumption, especially in France, policymakers might aim to level down recipients' housing choices, especially through the housing size. Hence, we assess policy efficiency in both situations (positively and negatively valued), while we add a situation with a zero valuation of the intensive margin effect (denoted as the indifferent situation). We report the results in [Table 5](#).

Table 5. Return on Investment Measures According to Different Valuation of Extensive and Intensive Margin Effects from a Policymaker Perspective

ψ	$\phi = 5k$				$\phi = 1k$			
	-2k	-1k	0k	1k	-2k	-1k	0k	1k
Covering Share	-0.62 (-26.1)	2.23 (-22.0)	5.08 (-20.7)	7.93 (-22.8)	-5.49 (-17.2)	-2.64 (-9.2)	0.22 (-4.1)	3.07 (-9.6)
Ceiling Value	-1.16 (4.2)	-2.33 (3.2)	-3.50 (2.7)	-4.67 (3.1)	0.84 (3.2)	-0.33 (1.7)	-1.50*** (0.5)	-2.67* (1.6)
Interest Rate	0.29 (2.1)	-0.72 (1.7)	-1.74 (1.5)	-2.75* (1.6)	0.88 (1.3)	-0.13 (0.7)	-1.15*** (0.3)	-2.16*** (0.7)
Maturity	-0.95 (1.9)	-0.48 (1.5)	-0.02 (1.3)	0.44 (1.4)	-1.73 (1.4)	-1.27* (0.8)	-0.80*** (0.3)	-0.34 (0.7)

Notes: Exploiting coefficients derived from the dose-response function ([Table 4](#)), we calculate return on investment for a cost-normalised increase of the overall IFL budget using [Equation 3](#) for the four sources of treatment variables. Our results can be interpreted as the monetary benefits from a 1 euro increase in the IFL budget cost from a policymaker perspective. As our return on investment measure depends on the marginal valuation of extensive margin effects (ϕ), price capitalization (ψ) and distortion of housing choices, we simulate different scenarios. The top panel (respectively bottom panel) corresponds to the situation in which an additional homeowner is valued at 10k (respectively 1k). Although price capitalization is likely to be negatively valued, we also assess scenarios with opposite signs according to our expectations. We report between parentheses the standard errors using the 500-iteration bootstrap procedure.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

*** $p < 0.01$, ** $p < 0.05$ * $p < 0.1$

Regardless of extensive and intensive margin valuations in credible scenarios, the return on investment of additional spending is not positive ([Table 5](#)). The return on investment is indeed at best non-significant, while it might be negative in some situations, that corresponds to a counter-productive effect of public spending. When valuing additional homeowners to 1k, one additional euro through raising the ceiling value induces a €2.7 loss when intensive margin effects are also sought by policymakers ($\phi = 1k$). The loss is however less important when policymakers are indifferent to the intensive margin effects ($\phi = 0k$). Remark that we find similar results when the additional public spending is caused by interest rate variation.

In conclusion, raising the subsidy regardless of the channel is at best inefficient, while

it might be counter-productive in specific situations as the return on investment is negative. The policy mostly modifies recipients' housing choices, and favours opportunistic behaviour, while it does not strengthen homeownership development.

6 Conclusion

The French IFL policy aims to increase the number of new homeowners through interest cuts for eligible households. We leverage spatial and temporal variation of treatment using a selection-on-observables approach to assess the effect of subsidy variation on policy objectives that relate either to extensive margin or intensive margin. Our GPS specification and regression adjustment involve, among other variables, spatial coordinates to prevent omitted spatial variables. From linear dose-response functions, we discuss the policy results for different simulations that differ according to the policymaker valuation of extensive and intensive margin effects to derive conclusions about policy return on investment.

We cannot reject the possibility that increasing policy expenditure on the IFL does not affect the number of new homeowners. Indeed, based on our identification strategy, we cannot exclude that increasing both policy control treatment variables affects tenure decisions at the individual level. However, we precisely estimate that the intensive margin effects exceed potential ones at the extensive margin even though it is the latter that is being targeted by policymakers. In addition, increasing the IFL subsidy causes demand to shift from existing to new housing, resulting from opportunistic behaviour. It turns out that the relevance of the IFL mainly depends on the valuation of intensive margin effects. In credible situations with a positive valuation of the extensive margin, the return on investment from the policymaker's perspective is negative, or at best null. Hence, regardless of the treatment variable under consideration, increasing public spending to support homeownership through the IFL appears to be inefficient.

Our paper leaves open questions for further research on the assessment of public support for homeownership. As housing market capitalization is related to housing supply and land availability, externalities produced by interest cuts are likely to depend on local characteristics. Assessing the structure of such heterogeneity is crucial for more precisely assessing the IFL policy according to the areas policymakers aim to favour. Finally, as supporting homeownership affects recipients' housing choices, it raises concerns about the impact of interest cuts on land consumption and policy interaction with policy objectives of reducing land development for housing development.

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Online Appendix (not for publication)

A Additional Descriptive Statistics

A.1 ABC Perimeter

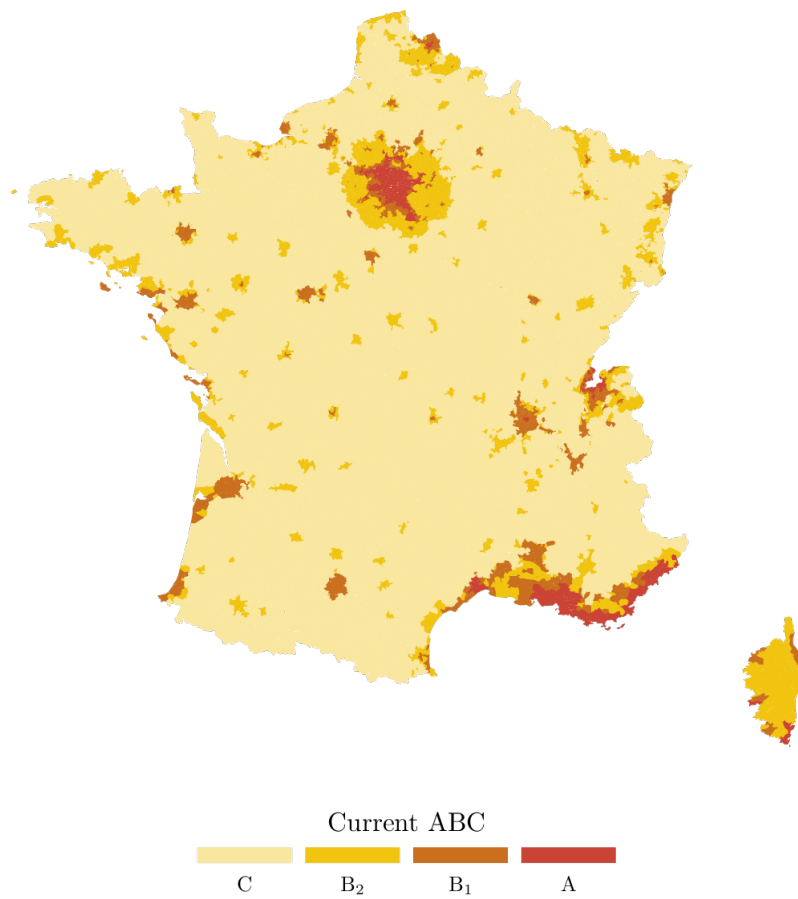


Figure A.1. Current ABC Zoning

A.2 Municipalities Pre-Treatment Characteristics

	N	Mean	Std Dev	Median	Q1	Q3	Min	Max
Density	26,819	1.926	8.058	0.497	0.244	1.142	0.007	259.982
CS1	26,819	2.91	3.86	1.59	0.18	4.00	0.00	55.00
CS2	26,819	4.34	2.82	3.93	2.57	5.65	0.00	31.25
CS3	26,819	5.81	4.47	4.88	2.79	7.84	0.00	38.46
CS4	26,819	13.50	5.60	13.33	9.62	17.13	0.00	45.00
CS5	26,819	15.87	4.83	15.87	12.98	18.64	0.00	60.14
CS6	26,819	15.43	6.23	15.00	11.11	19.21	0.00	55.00
CS7	26,819	29.92	9.14	29.07	23.65	35.38	0.00	87.50
Price	26,819	153,108	68,002	141,975	110,323	181,032	20,518	2,261,166
Price per m^2 (2010–2013)	26,819	1,608.7	691.8	1,471.0	1,181.4	1,855.4	159.3	19,306.5
Neigh Price per m^2	26,819	1,561.0	724.0	1,420.2	1,123.6	1,807.3	0.0	35,686.1
New Housing (2010–2013)	26,819	42	227	9	4	25	1	15,748
Median Income (2013)	26,819	19,954	3,399	19,432	17,774	21,546	8,774	47,316
Longitude (WGS 84)	26,819	653,319	187,946	653,382	511,822	802,857	124,073	1,072,432
Latitude (WGS 84)	26,819	6,651,138	243,230	6,677,060	6,448,774	6,858,734	6,139,677	7,108,696

Notes: The average density of the municipalities used to estimate the GPS is 193 inhabitants per kilometer square. Our sample is composed of 26,819 observations. CS1 corresponds to share of socio-professional categories within the municipality. 1 corresponds to farmers, 2 to artisans and merchants, 3 to managers, 4 to intermediate professions, 5 to employees, 6 to labor works, 7 to retired.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

A.3 Descriptive Statistics for the Outcomes

Table A.3. Descriptive Statistics for the Outcomes

	N	Mean	Std Dev	Median	Q1	Q3	Min	Max
A								
FTO	1,874	258.7	614.6	118.0	48.0	48.0	1.0	12,760.0
Price (Transaction)	1,874	443,572	426,877	320,763	261,133	261,133	107,760	5,999,507
Surface (Transaction)	1,874	80	17	78	68	68	39	177
Unit. Price (Transaction)	1,874	7,279	11,407	4,416	3,470	3,470	1,197	238,899
Recipients	1,874	46.8	102.6	17.0	5.0	5.0	1.0	2,182.0
Price (IFL)	1,874	281,426	67,118	273,749	232,633	232,633	107,000	660,000
Surface (IFL)	1,874	81	25	75	62	62	22	271
Unit. Price (IFL)	1,874	3,662	1,096	3,407	3,003	3,003	641	10,897
Cost	1,874	19,068	5,295	18,568	15,694	15,694	2,264	40,806
B ₁								
FTO	3,295	122.4	348.7	48.0	24.0	24.0	1.0	8,268.0
Price (Transaction)	3,295	309,657	263,593	254,787	208,695	208,695	111,964	7,386,864
Surface (Transaction)	3,295	90	17	89	79	79	39	175
Unit. Price (Transaction)	3,295	4,174	5,056	3,095	2,536	2,536	1,442	119,469
Recipients	3,295	18.7	38.4	8.0	3.0	3.0	1.0	848.0
Price (IFL)	3,295	238,583	53,601	232,025	202,959	202,959	68,441	620,610
Surface (IFL)	3,295	95	23	95	80	80	30	280
Unit. Price (IFL)	3,295	2,595	613	2,499	2,197	2,197	615	10,753
Cost	3,295	17,445	4,734	17,189	14,408	14,408	2,422	41,496
B ₂								
FTO	6,200	55.2	129.4	25.0	12.0	12.0	1.0	2,572.0
Price (Transaction)	6,200	213,863	201,699	186,010	154,125	154,125	20,000	6,297,033
Surface (Transaction)	6,200	96	16	95	86	86	38	191
Unit. Price (Transaction)	6,200	2,630	3,528	2,117	1,713	1,713	345	125,116
Recipients	6,200	8.2	11.5	4.0	2.0	2.0	1.0	127.0
Price (IFL)	6,200	207,007	41,762	200,911	179,365	179,365	60,691	479,954
Surface (IFL)	6,200	103	21	101	93	93	1	500
Unit. Price (IFL)	6,200	2,117	3,142	1,978	1,766	1,766	372	172,197
Cost	6,200	11,906	5,642	10,576	7,370	7,370	995	47,635
C								
FTO	29,463	17.5	23.1	11.0	6.0	6.0	1.0	424.0
Price (Transaction)	29,463	161,697	137,386	144,000	114,000	114,000	12,000	6,258,743
Surface (Transaction)	29,463	100	18	99	90	90	20	400
Unit. Price (Transaction)	29,463	1,849	2,728	1,534	1,229	1,229	138	169,770
Recipients	29,463	3.7	5.7	2.0	1.0	1.0	1.0	169.0
Price (IFL)	29,463	181,554	37,411	178,500	157,928	157,928	40,000	492,888
Surface (IFL)	29,463	108	22	104	95	95	1	700
Unit. Price (IFL)	29,463	1,769	2,769	1,706	1,504	1,504	165	243,577
Cost	29,463	9,851	5,117	8,489	5,729	5,729	322	41,061

B IFL Design

B.1 Cost Calculation

The monetary benefit of IFL for the subsidized first-time owner is equal to the cost for the government, and without extensive and intensive margins, the IFL policy is just a transfer. Consider a loan of total value V_b for a loan duration d at a yearly interest rate of r . For each due date, t , the new homeowner reimburses a fixed payment m . The remaining capital to reimburse at the end of the year is:

$$X_t = X_{t-1} - m + rX_{t-1} = (1 + r) X_{t-1} - m \quad (11)$$

Then after calculation, we obtain, using the condition $X_0 = V_b$

$$X_t = (1 + r)^t \left[V_b - \frac{m}{r} \right] + \frac{m}{r} \quad (12)$$

Thus, we estimate the monthly payment using $X_D = 0$, corresponding to the loan maturity. Hence, we obtain:

$$m = \frac{rX_0}{1 - (1 + r)^{-D}} \quad (13)$$

yielding an overall cost for the household to

$$C = \sum_{k=1}^D m - V_b = \left[\frac{D r}{1 - (1 + r)^{-D}} - 1 \right] V_b \quad (14)$$

C Additional Results from Models Estimations

C.1 Estimated Spline Functions for the GPS specification

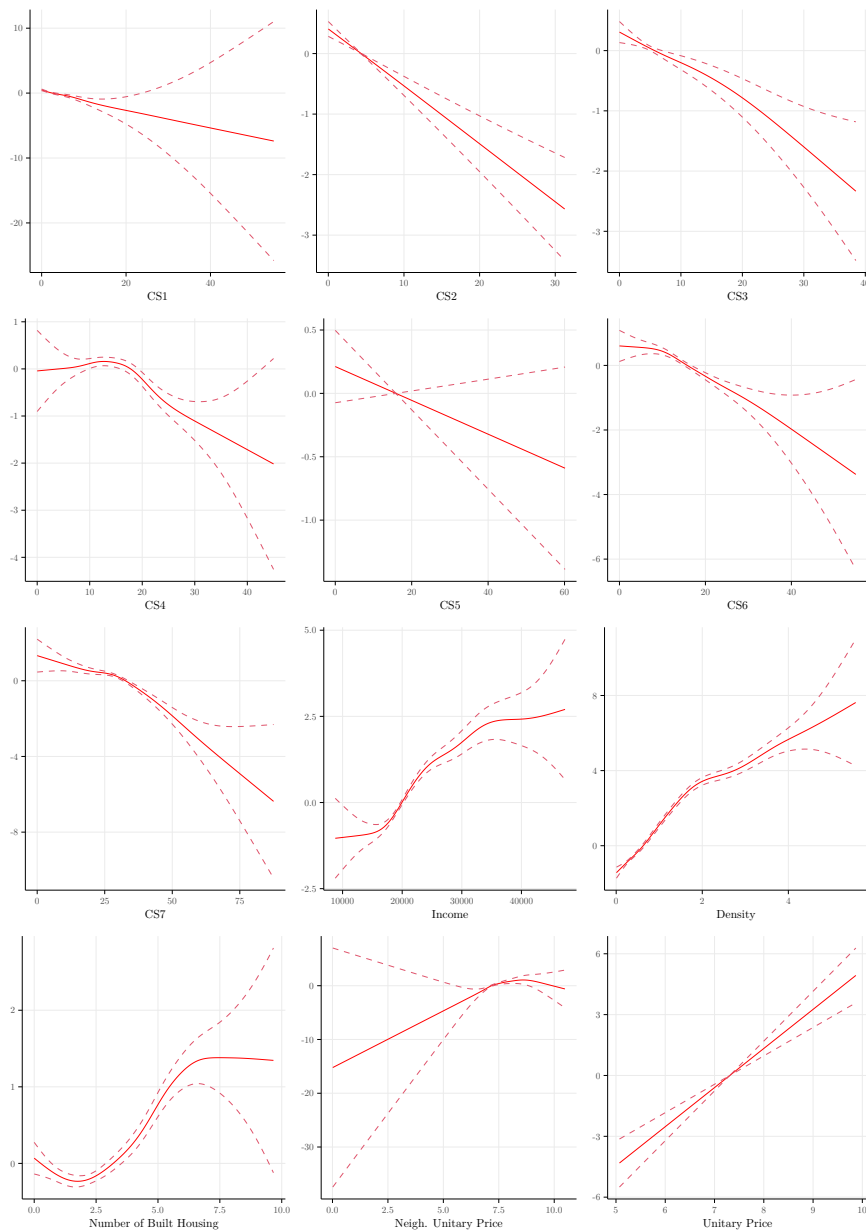


Figure C.1. Contribution for 1D-variable in the GPS Estimation

Notes: For each continuous covariate, we report the functional form in the GPS estimation following the endogenous shrinkage procedure to set the effective degree of freedom. In addition, we report the confidence interval for a 95% level. We exploit the `gam` function from the `mgcv` package.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

C.2 Spatial Smoothing Splines for GPS Estimation

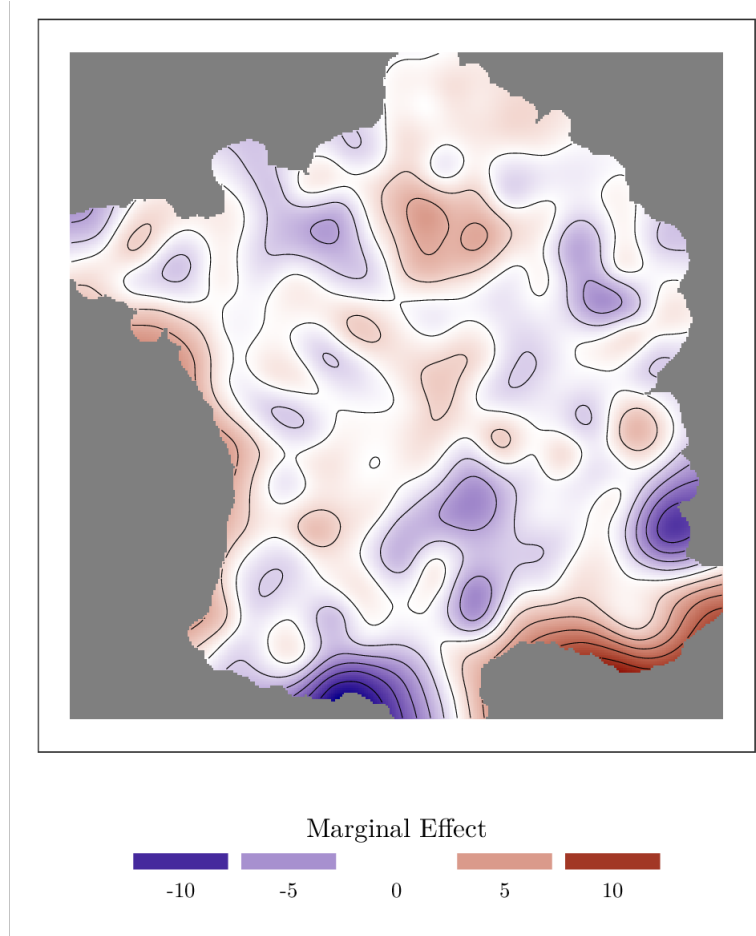


Figure C.2. Spatial Smoothing Function for the GPS Estimation Based On Municipality Coordinates

Notes: We report the spatial smoothing function for the GPS estimation, using bi-variate additive splines. Spline parameters are endogenously shrunk using restricted maximum likelihood approach. The maximum degree of freedom is set to 200. Red (respectively blue) values indicate that the outcome is locally higher than the average. We exploit the `gam` function from the `mgcv` package.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

C.3 Main Results With Maximum Degree of Freedom set to 100 for Spatial Smoothing

Table C.3. OLS coefficients for policy primitives from dose-response functions

	<i>Outcome variables from ...</i>								
	Tax	Transaction Data			IFL files				
	N	V	S	P	$\tilde{N}$	$\tilde{V}$	$\tilde{S}$	$\tilde{P}$	$\tilde{C}$
Covering Share	0.009 (0.120)	0.010 (0.021)	-0.056 (0.049)	-0.017 (0.051)	0.031 (0.075)	0.020 (0.017)	-0.008 (0.017)	0.013 (0.016)	0.047*** (0.014)
Ceiling Value	-0.054 (0.127)	-0.007 (0.019)	0.035 (0.039)	0.039 (0.040)	0.040 (0.089)	-0.010 (0.025)	0.015 (0.023)	0.008 (0.014)	0.016 (0.012)
Interest Rate	-0.013 (0.034)	0.001 (0.005)	-0.004 (0.008)	0.002 (0.008)	0.024 (0.026)	-0.000 (0.007)	0.000 (0.007)	0.002 (0.004)	0.006 (0.004)
Loan Maturity	0.019 (0.074)	0.007 (0.017)	-0.034 (0.027)	-0.036 (0.031)	0.016 (0.039)	0.008 (0.009)	-0.012 (0.009)	-0.004 (0.007)	0.017** (0.009)
Constant	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.261** (0.125)	0.025 (0.038)	0.001 (0.039)	0.025 (0.029)	-0.006 (0.030)
R ²	0.809	0.201	0.318	0.415	0.204	0.129	0.138	0.355	0.834
Adj. R ²	0.784	0.097	0.231	0.339	0.101	0.016	0.027	0.271	0.812
N	36	36	36	36	36	36	36	36	36

Notes: For the nine outcomes Y in columns, the table reports the β_a^Y coefficients associated to each primitive in rows. They are estimated from dose-response functions of [Equation 5](#). N accounts for the number of new homeowners, V for housing value, S for surface, and P for unitary housing price. The variables with a $\tilde{\cdot}$ are the same variables computed for IFL recipients, $\tilde{C}$ is the IFL cost. The unit of observation is the bilateral combination of four ABC zones for the three periods of interest. The maximum degree of freedom for the spatial smoothing in the specifications of both propensity score and outcomes is set to 100. Standards errors in parentheses are estimated by bootstrap with 500 iterations. ATEs for tax and transaction data are weighted according to the inverse of their bootstrapped standard errors, ATT for IFL variables are additionally weighted according to the number of municipalities receiving the considered treatment levels.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

*** $p < 0.01$, ** $p < 0.05$ * $p < 0.1$

C.4 Policy Relevant Treatment Effects (ATE)

Area	2015				Treatement (ATE) 2016–2017				2018–2019			
	A	B ₁	B ₂	C	A	B ₁	B ₂	C	A	B ₁	B ₂	C
	<i>N_g</i> : Number of FTO											
A	-	0.242	0.451	0.469	-	-1.180	-0.359	-0.508	-	1.168	1.026	0.979
	-	(1.231)	(1.213)	(1.219)	-	(1.032)	(0.953)	(0.950)	-	(1.202)	(1.167)	(1.164)
B ₁	-0.242	-	0.209	0.227	1.180	-	0.821**	0.672*	-1.168	-	-0.142	-0.189
	(1.231)	-	(0.185)	(0.174)	(1.032)	-	(0.393)	(0.389)	(1.202)	-	(0.297)	(0.295)
B ₂	-0.451	-0.209	-	0.018	0.359	-0.821**	-	-0.149**	-1.026	0.142	-	-0.048
	(1.213)	(0.185)	-	(0.067)	(0.953)	(0.393)	-	(0.061)	(1.167)	(0.297)	-	(0.053)
C	-0.469	-0.227	-0.018	-	0.508	-0.672*	0.149**	-	-0.979	0.189	0.048	-
	(1.219)	(0.174)	(0.067)	-	(0.950)	(0.389)	(0.061)	-	(1.164)	(0.295)	(0.053)	-
<i>V_g</i> : Average Housing Price (Overall Transaction)												
A	-	-0.431	-0.129	-0.199	-	0.446	0.488	0.456	-	0.294	-0.202	-0.248
	-	(0.784)	(0.730)	(0.729)	-	(0.583)	(0.564)	(0.562)	-	(1.195)	(1.184)	(1.187)
B ₁	0.431	-	0.301	0.232	-0.446	-	0.043	0.010	-0.294	-	-0.496***	-0.542***
	(0.784)	-	(0.255)	(0.254)	(0.583)	-	(0.167)	(0.169)	(1.195)	-	(0.142)	(0.135)
B ₂	0.129	-0.301	-	-0.070***	-0.488	-0.043	-	-0.033	0.202	0.496***	-	-0.046
	(0.730)	(0.255)	-	(0.021)	(0.564)	(0.167)	-	(0.021)	(1.184)	(0.142)	-	(0.036)
C	0.199	-0.232	0.070***	-	-0.456	-0.010	0.033	-	0.248	0.542***	0.046	-
	(0.729)	(0.254)	(0.021)	-	(0.562)	(0.169)	(0.021)	-	(1.187)	(0.135)	(0.036)	-
<i>Q_g</i> : Average Housing Size (Overall Transaction)												
A	-	-0.081	0.132	0.113	-	0.177	0.147	0.168	-	0.270	0.128	0.158
	-	(0.294)	(0.286)	(0.284)	-	(0.324)	(0.321)	(0.321)	-	(0.293)	(0.289)	(0.290)
B ₁	0.081	-	0.213***	0.194***	-0.177	-	-0.030	-0.010	-0.270	-	-0.142**	-0.112*
	(0.294)	-	(0.074)	(0.072)	(0.324)	-	(0.054)	(0.053)	(0.293)	-	(0.065)	(0.065)
B ₂	-0.132	-0.213***	-	-0.019	-0.147	0.030	-	0.021	-0.128	0.142**	-	0.030**
	(0.286)	(0.074)	-	(0.019)	(0.321)	(0.054)	-	(0.015)	(0.289)	(0.065)	-	(0.014)
C	-0.113	-0.194***	0.019	-	-0.168	0.010	-0.021	-	-0.158	0.112*	-0.030**	-
	(0.284)	(0.072)	(0.019)	-	(0.321)	(0.053)	(0.015)	-	(0.290)	(0.065)	(0.014)	-
<i>P_g</i> : Average Housing Price per m ² (Overall Transaction)												
A	-	-0.116	0.092	0.061	-	0.209	-0.021	-0.103	-	0.259	-0.146	-0.227
	-	(0.867)	(0.852)	(0.850)	-	(0.588)	(0.562)	(0.561)	-	(1.087)	(1.080)	(1.081)
B ₁	0.116	-	0.209	0.177	-0.209	-	-0.229	-0.312**	-0.259	-	-0.404***	-0.485***
	(0.867)	-	(0.208)	(0.203)	(0.588)	-	(0.154)	(0.156)	(1.087)	-	(0.137)	(0.133)
B ₂	-0.092	-0.209	-	-0.031	0.021	0.229	-	-0.082***	0.146	0.404***	-	-0.081**
	(0.852)	(0.208)	-	(0.037)	(0.562)	(0.154)	-	(0.022)	(1.080)	(0.137)	-	(0.032)
C	-0.061	-0.177	0.031	-	0.103	0.312**	0.082***	-	0.227	0.485***	0.081**	-
	(0.850)	(0.203)	(0.037)	-	(0.561)	(0.156)	(0.022)	-	(1.081)	(0.133)	(0.032)	-

Notes: We report the bilateral combinations effect for ATE type estimator. The four panels correspond to the four outcome concerned by ATE estimation and derived from fiscal data. Then, we have three main columns that represent the stable period for the IFL scheme, with four subcolumns related to the ABC classification. In rows, we have again the levels contained in the ABC classification. Hence, the bilateral combinations are reported for each intersection, and must be understood as “if (rows) have received (cols), difference in outcome would be (results)”. We also report in brackets the standard errors obtained with a bootstrap procedure with 500 iterations.

Sources: Authors’ Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.*** p < 0.01, ** p < 0.05 * p < 0.1

C.5 Policy Relevant Treatment Effects (ATT)

Area	2015				<i>Treatment (ATT)</i> 2016–2017				2018–2019			
	A	B ₁	B ₂	C	A	B ₁	B ₂	C	A	B ₁	B ₂	C
	$\tilde{N}_g$: Number of IFL											
A	-	-0.999*** (0.266)	-0.804* (0.434)	-0.938 (0.616)	-	-0.289 (0.180)	-0.871** (0.408)	-2.548*** (0.600)	-	-0.095 (0.230)	-0.668 (0.540)	-1.933* (1.024)
B ₁	3.446** (1.587)	-	-0.196** (0.079)	-0.290 (0.179)	0.275 (1.450)	-	-0.205** (0.090)	-0.547*** (0.172)	-1.715 (1.209)	-	-0.210** (0.089)	-0.576* (0.297)
B ₂	3.809* (2.036)	-0.145 (0.158)	-	-0.163*** (0.052)	0.185 (2.103)	-0.204 (0.186)	-	-0.233*** (0.051)	-0.279 (1.940)	-0.124 (0.179)	-	-0.261*** (0.059)
C	5.385* (3.138)	0.275 (0.695)	-0.009 (0.100)	-	2.255 (3.344)	-0.095 (1.090)	0.002 (0.138)	-	0.670 (3.067)	-1.472* (0.814)	0.066 (0.131)	-
$\tilde{V}_g$: Average Housing Price (Subsidized Housing)												
A	-	-0.028 (0.054)	-0.191** (0.082)	-0.076 (0.103)	-	-0.022 (0.027)	-0.011 (0.054)	-0.060 (0.096)	-	-0.069* (0.036)	-0.161*** (0.060)	-0.046 (0.085)
B ₁	-0.801*** (0.297)	-	-0.043** (0.017)	-0.034 (0.032)	-0.118 (0.229)	-	-0.050*** (0.013)	-0.036 (0.025)	-0.293 (0.273)	-	-0.048*** (0.014)	-0.048* (0.028)
B ₂	-1.000** (0.391)	0.096*** (0.030)	-	-0.030*** (0.011)	-0.163 (0.313)	0.054 (0.035)	-	-0.019** (0.008)	-0.090 (0.378)	0.103*** (0.028)	-	-0.032*** (0.009)
C	-1.508** (0.613)	0.160 (0.123)	0.075*** (0.022)	-	0.222 (0.514)	0.100 (0.154)	0.033 (0.044)	-	0.207 (0.595)	0.307** (0.124)	0.037 (0.038)	-
$\tilde{Q}_g$: Average Housing Size (Subsidized Housing)												
A	-	-0.109 (0.087)	0.167* (0.090)	0.654*** (0.160)	-	-0.073* (0.041)	0.077 (0.066)	0.242** (0.098)	-	-0.083* (0.049)	0.051 (0.069)	0.350** (0.144)
B ₁	-1.027** (0.457)	-	0.022 (0.027)	0.125*** (0.045)	-0.409 (0.290)	-	0.011 (0.018)	0.057** (0.027)	-0.448 (0.319)	-	0.023 (0.017)	0.117*** (0.045)
B ₂	-1.062* (0.582)	0.026 (0.039)	-	0.014 (0.013)	-0.996** (0.408)	0.089* (0.053)	-	0.010 (0.008)	-0.734 (0.469)	0.019 (0.036)	-	-0.001 (0.014)
C	-1.997** (0.894)	0.216 (0.155)	0.054** (0.027)	-	-2.287*** (0.695)	0.473 (0.288)	0.012 (0.058)	-	-0.941 (0.778)	0.262* (0.158)	-0.046 (0.058)	-
$\tilde{P}_g$: Average Housing Price per m ² (Subsidized Housing)												
A	-	0.100 (0.087)	-0.355*** (0.101)	-0.460*** (0.135)	-	0.027 (0.037)	-0.094* (0.052)	-0.302*** (0.093)	-	0.017 (0.042)	-0.214*** (0.080)	-0.396*** (0.124)
B ₁	0.231 (0.370)	-	-0.071*** (0.022)	-0.128*** (0.039)	0.382 (0.263)	-	-0.063*** (0.014)	-0.093*** (0.023)	0.212 (0.233)	-	-0.069*** (0.018)	-0.165*** (0.039)
B ₂	0.116 (0.467)	0.037 (0.039)	-	-0.044*** (0.014)	0.978** (0.482)	-0.042 (0.070)	-	-0.026*** (0.007)	0.624* (0.332)	0.077*** (0.025)	-	-0.032*** (0.012)
C	0.484 (0.705)	-0.046 (0.149)	0.018 (0.030)	-	1.822** (0.836)	-0.366 (0.340)	0.026 (0.026)	-	1.219** (0.548)	0.079 (0.119)	0.084 (0.067)	-
$\tilde{C}_g$: Average Cost per IFL												
A	-	-0.125 (0.079)	-0.516*** (0.162)	-1.514*** (0.393)	-	-0.056 (0.043)	-0.226** (0.096)	-0.173 (0.179)	-	-0.110** (0.049)	-1.096*** (0.115)	-0.745*** (0.148)
B ₁	-0.096 (0.430)	-	-0.527*** (0.042)	-1.148*** (0.106)	0.155 (0.226)	-	-0.200*** (0.025)	-0.266*** (0.045)	-0.229 (0.347)	-	-0.874*** (0.019)	-0.922*** (0.043)
B ₂	0.245 (0.560)	0.509*** (0.056)	-	-0.490*** (0.030)	-0.151 (0.364)	0.144*** (0.037)	-	-0.110*** (0.014)	0.664 (0.580)	0.876*** (0.059)	-	-0.126*** (0.017)
C	0.532 (0.908)	0.905*** (0.224)	0.555*** (0.086)	-	-0.432 (0.628)	0.508** (0.222)	0.139*** (0.033)	-	1.197 (0.979)	0.976*** (0.232)	0.140*** (0.039)	-

Notes: We report the bilateral combinations effect for ATT type estimator. The five panels correspond to the five outcomes concerned by ATT estimation and derived from recipients' files. Then, we have three main columns that represent the stable period for the IFL scheme, with four subcolumns related to the ABC classification. In rows, we have again the levels contained in the ABC classification. Hence, the bilateral combinations are reported for each intersection, and must be understood as "if (rows) have received (cols), difference in outcome would be (results)". We also report in brackets the standard errors obtained with a bootstrap procedure with 500 iterations.

C.6 Partial Plots (1)

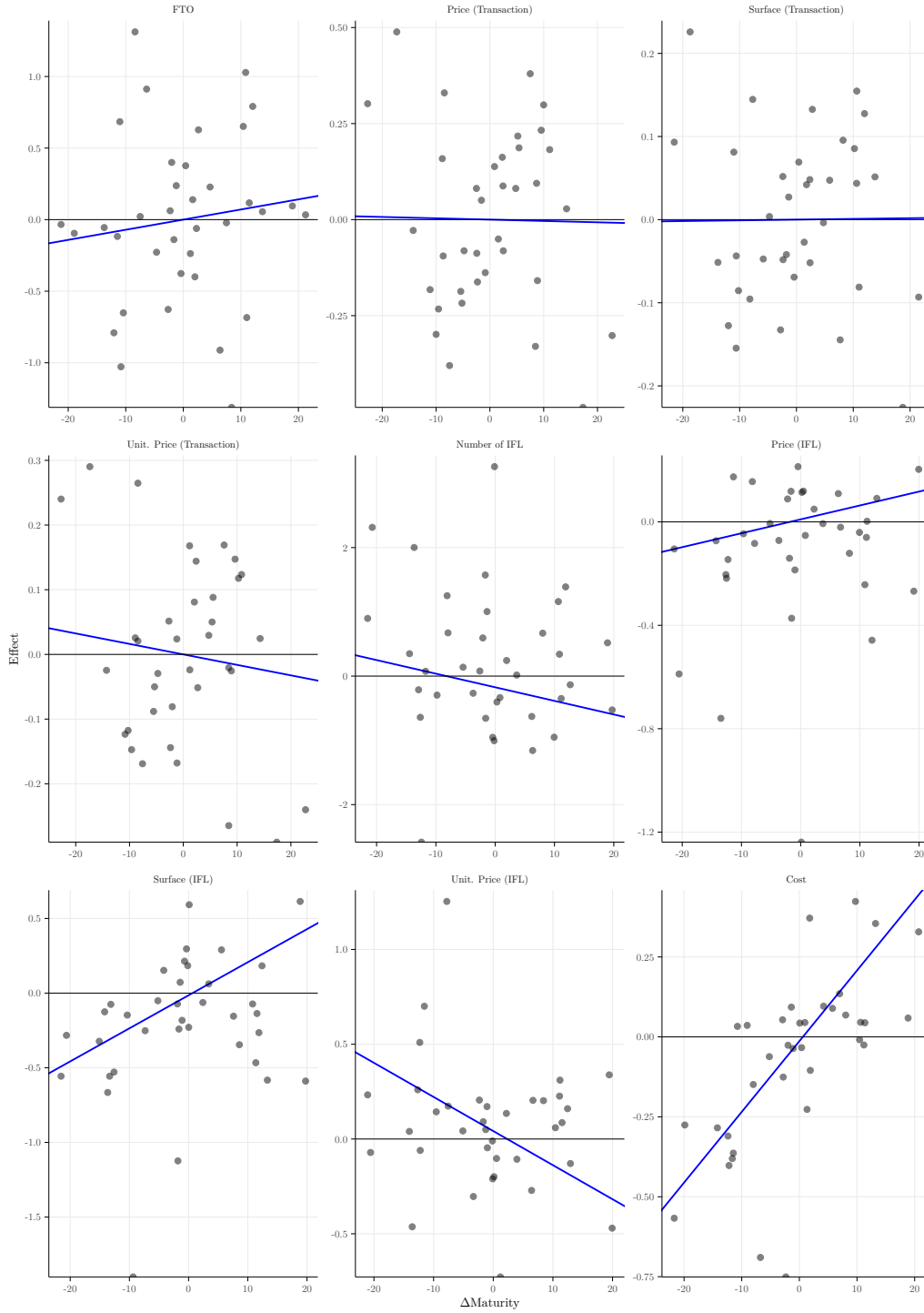


Figure C.6. Dose-Response Plots

Notes: We report the partial plot for dose-response function. The nine partial plot corresponds to our nine selected outcomes, while the x-axis represents a variation in primitive data sources. The observation unit is the bilateral combinations of treatment (either ATE or ATT). The regression is performed using OLS.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

C.7 Partial Plots (2)

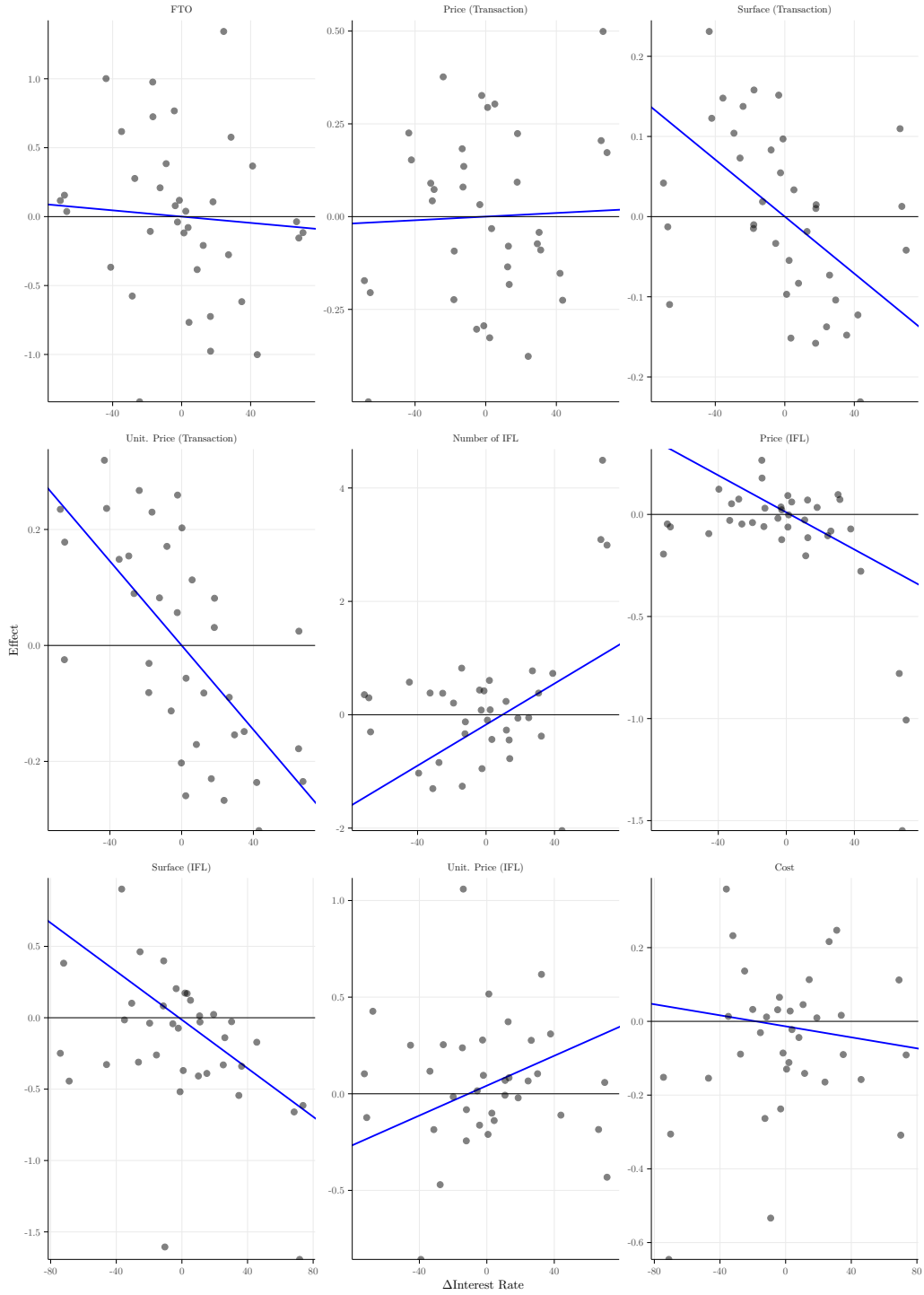


Figure C.7. Dose-Response Plots

Notes: We report the partial plot for dose-response function. The nine partial plot corresponds to our nine selected outcomes, while the x-axis represents a variation in primitive data sources. The observation unit is the bilateral combinations of treatment (either ATE or ATT). The regression is performed using OLS.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

C.8 Partial Plots (3)

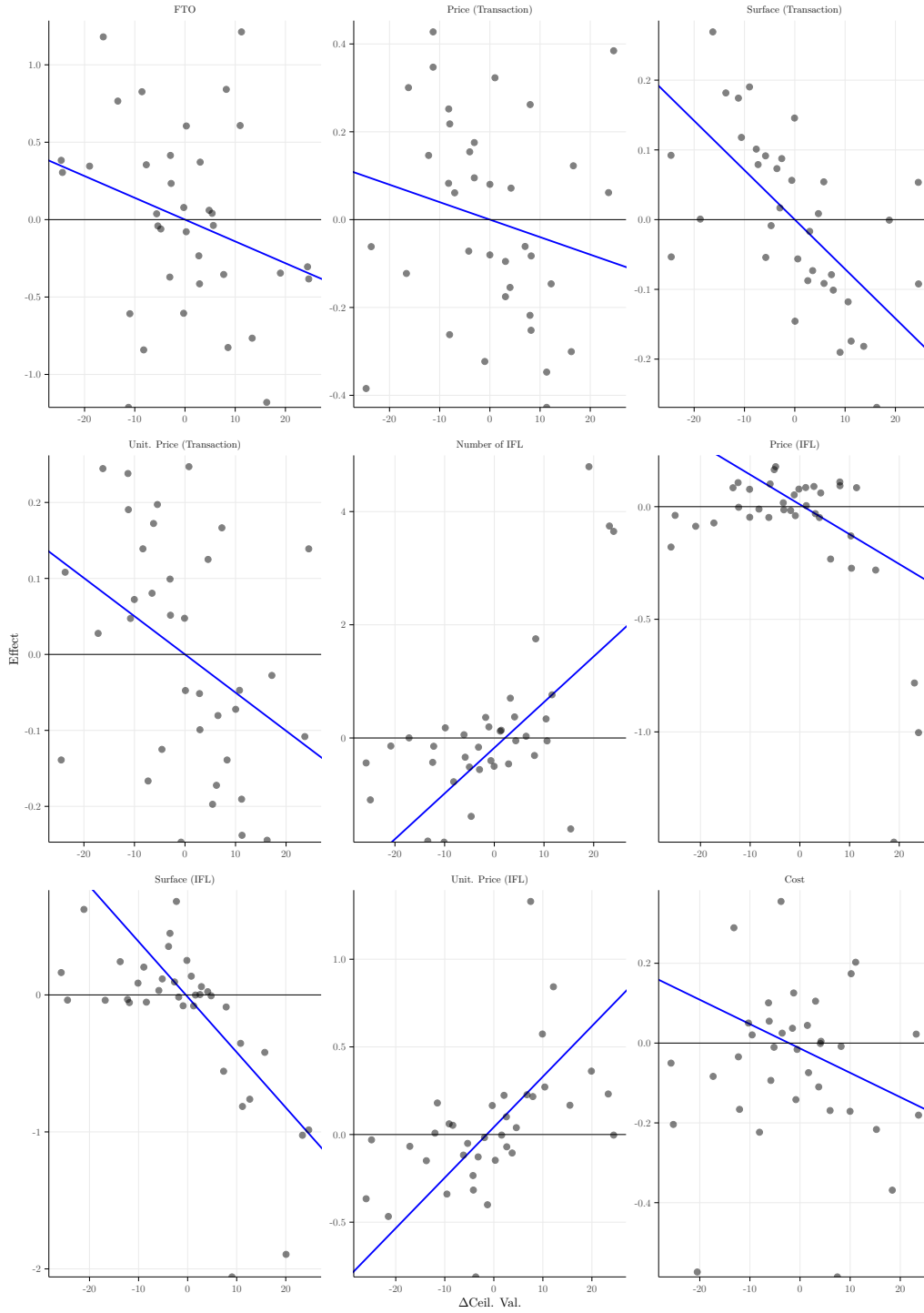


Figure C.8. Dose-Response Plots

Notes: We report the partial plot for dose-response function. The nine partial plot corresponds to our nine selected outcomes, while the x-axis represents a variation in primitive data sources. The observation unit is the bilateral combinations of treatment (either ATE or ATT). The regression is performed using OLS.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

C.9 Partial Plots (4)

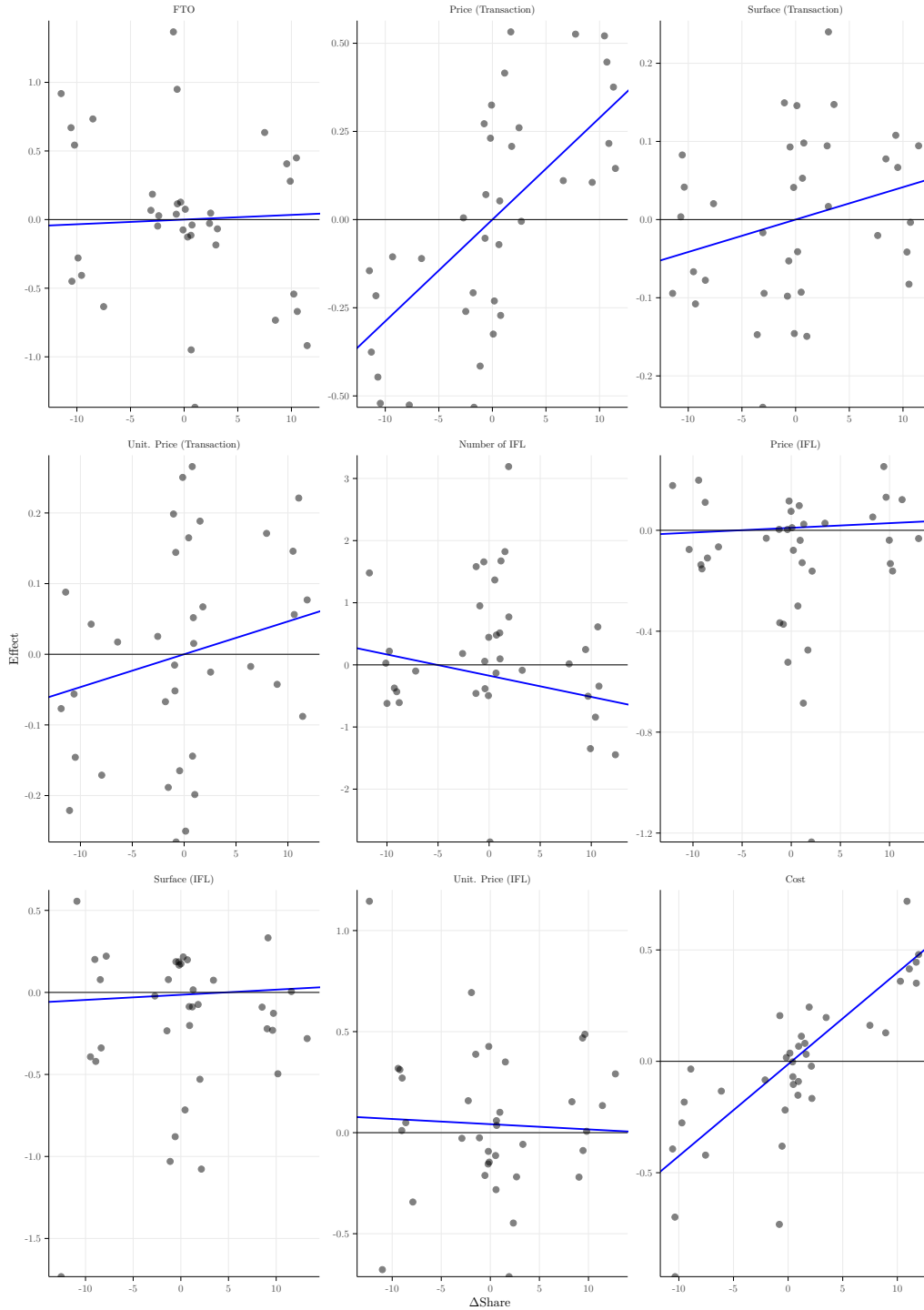


Figure C.9. Dose-Response Plots

Notes: We report the partial plot for dose-response function. The nine partial plot corresponds to our nine selected outcomes, while the x-axis represents a variation in primitive data sources. The observation unit is the bilateral combinations of treatment (either ATE or ATT). The regression is performed using OLS.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D Difference-in-Difference Results

D.1 Design of the Natural Experiments

We estimate causal effects of IFL policy based on alternative identification strategy by leveraging a natural experiment that occurred in January 2018. Indeed, a major reform affected significantly the covering share of two areas from the ABC zoning (namely the B₂ and C tier) whereas it remains unaffected for two other areas (namely the A and the B₁).

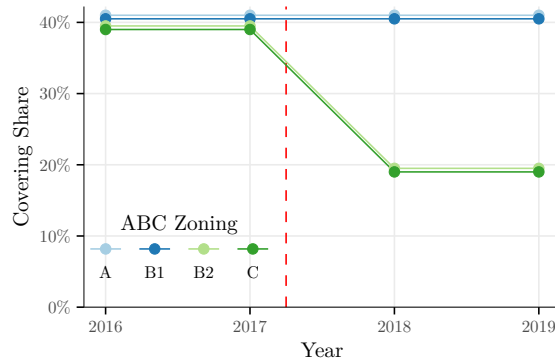


Figure D.1. Covering Share per ABC Areas According to Time

Notes: We represent the coverage share per ABC tier from 2016 to 2019. Green lines corresponds respectively to B₂ and C areas that experience a cut in the covering share that has been introduced in January 2018. The covering share falls from 40% to 20%. In the meantime, areas represented in blue lines does not experience any change of covering share in January 2018.

As the ABC zoning is endogenous, we adopt a difference-in-difference approach and consider each bilateral combination of treatment level (for instance the A-B₂), including bilateral combinations of areas with no difference in treatment as placebo test (for instance the A-B₁ combinations). We then estimate the following equation

$$Y_{jt} = \alpha \cdot \mathbb{1}_{t \geq 2018} + \beta \cdot \mathbb{1}_{j \in 40^-} + \gamma \cdot \mathbb{1}_{t \geq 2018} \times \mathbb{1}_{j \in 40^-} + \varepsilon_{jt} \quad (15)$$

with Y_{jt} outcome of interest for group j at time t , $\mathbb{1}_{j \in 40^-}$ indicating whether the group j is concerned by the downgrade of the covering share and ε_{jt} the error term. The parameter of interest γ captures the effect of reducing IFL intensity through the covering share. We estimate this equation by OLS with clustered standard errors at the level of commuting zones. We report results in the following subsections, considering the six possibilities under investigation.

D.2 DiD Estimation for B₁ and B₂ Areas

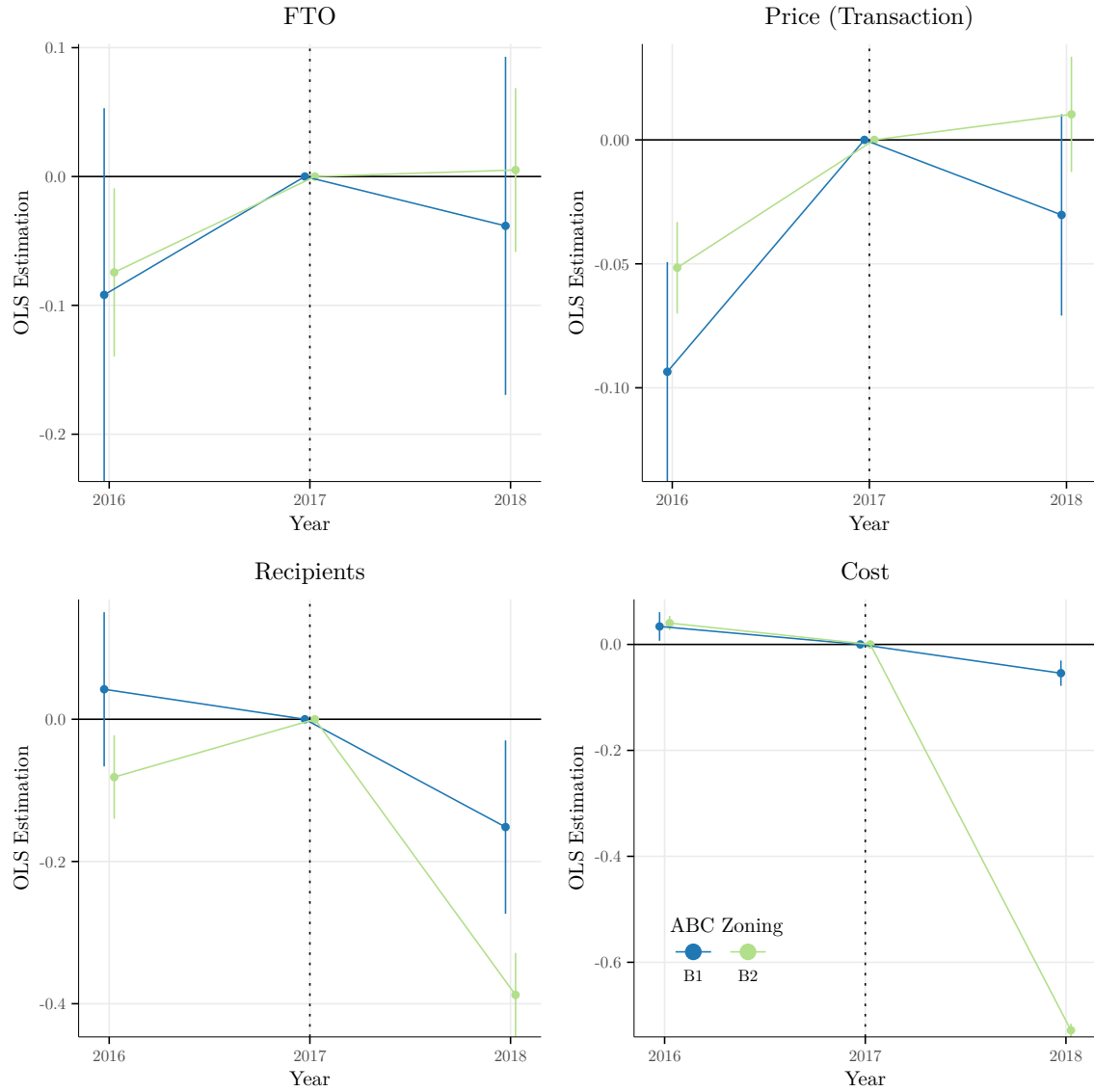


Figure D.2. OLS Results from DiD Identification Strategy restricted to B₁ and B₂ Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to B₁ and B₂ municipalities. B₂ municipalities experience a cut in the coverage share that occurred in January 2018. Conversely, B₁ municipalities do not experience any cut in the covering share. Our unit of observation is the municipality. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D.3 DiD Estimation for B₁ and C Areas

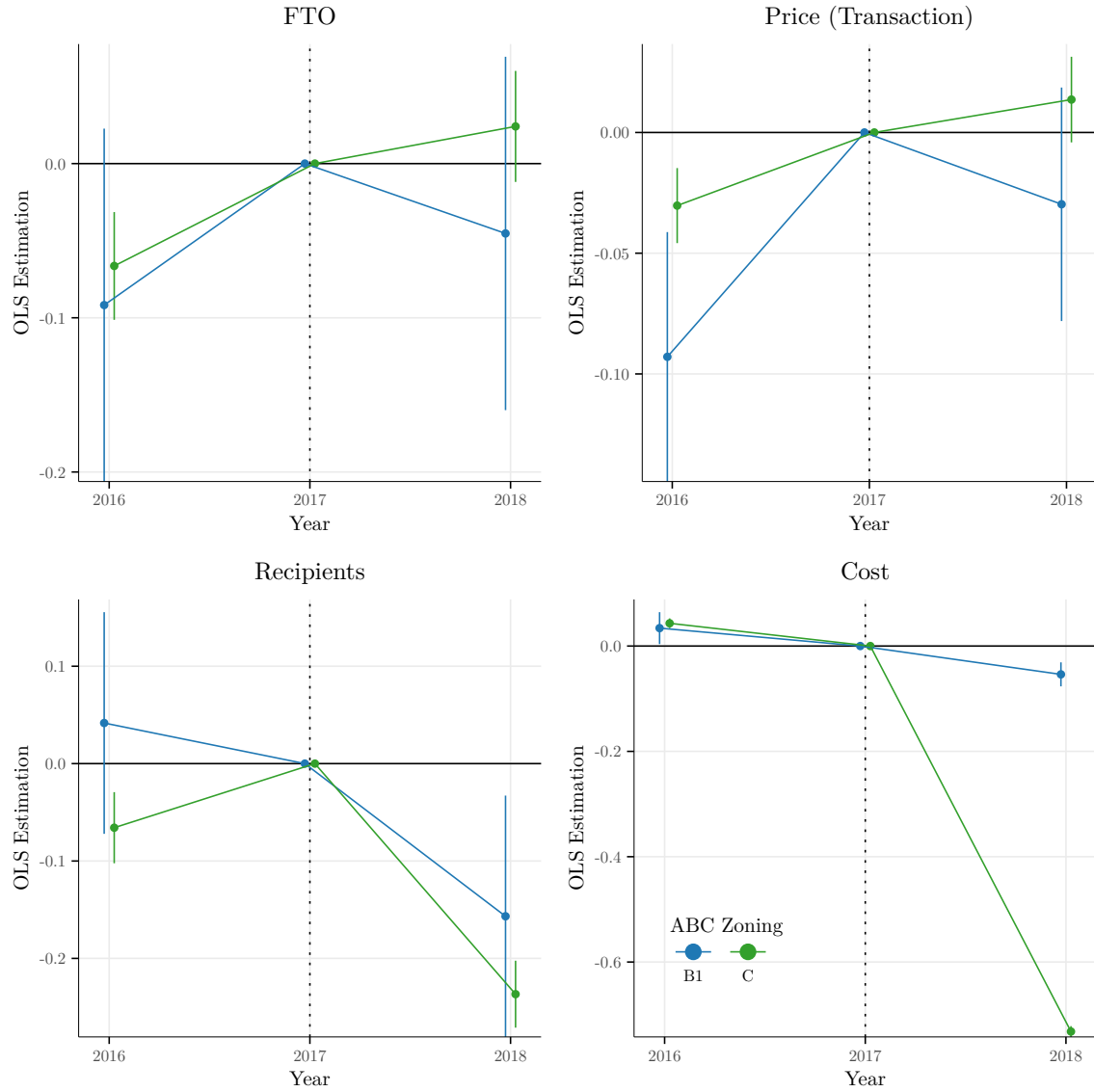


Figure D.3. OLS Results from DiD Identification Strategy restricted to B₁ and C Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to B₁ and C municipalities. C municipalities experience a cut in the coverage share that occurred in January 2018. Conversely, B₁ municipalities do not experience any cut in the covering share. Our unit of observation is the municipality. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D.4 DiD Estimation for A and B₂ Areas

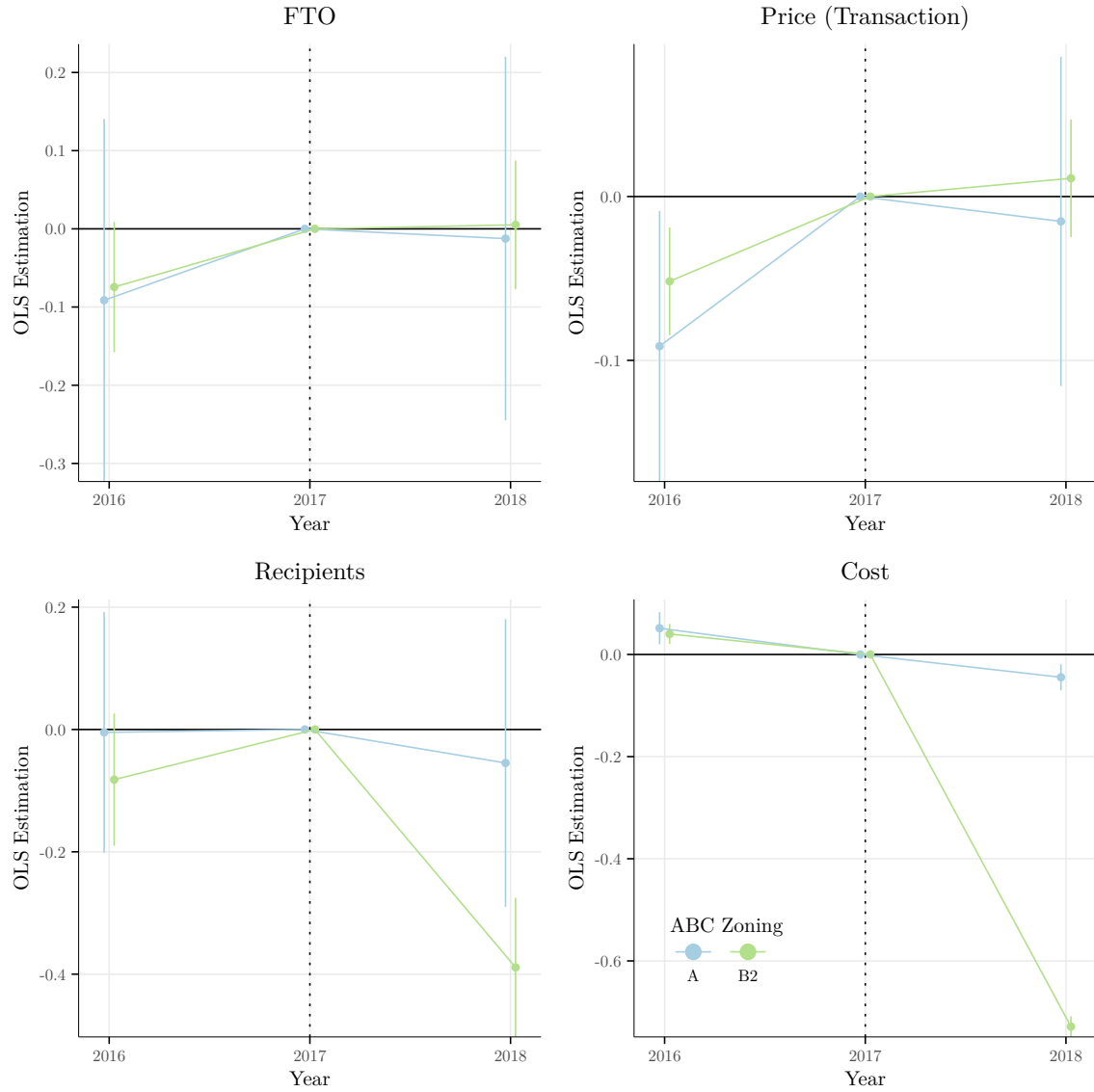


Figure D.4. OLS Results from DiD Identification Strategy restricted to A and B₂ Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to A and B₂ municipalities. B₂ municipalities experience a cut in the coverage share that occurred in January 2018. Conversely, A municipalities do not experience any cut in the covering share. Our unit of observation is the municipality. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D.5 DiD Estimation for A and C Areas

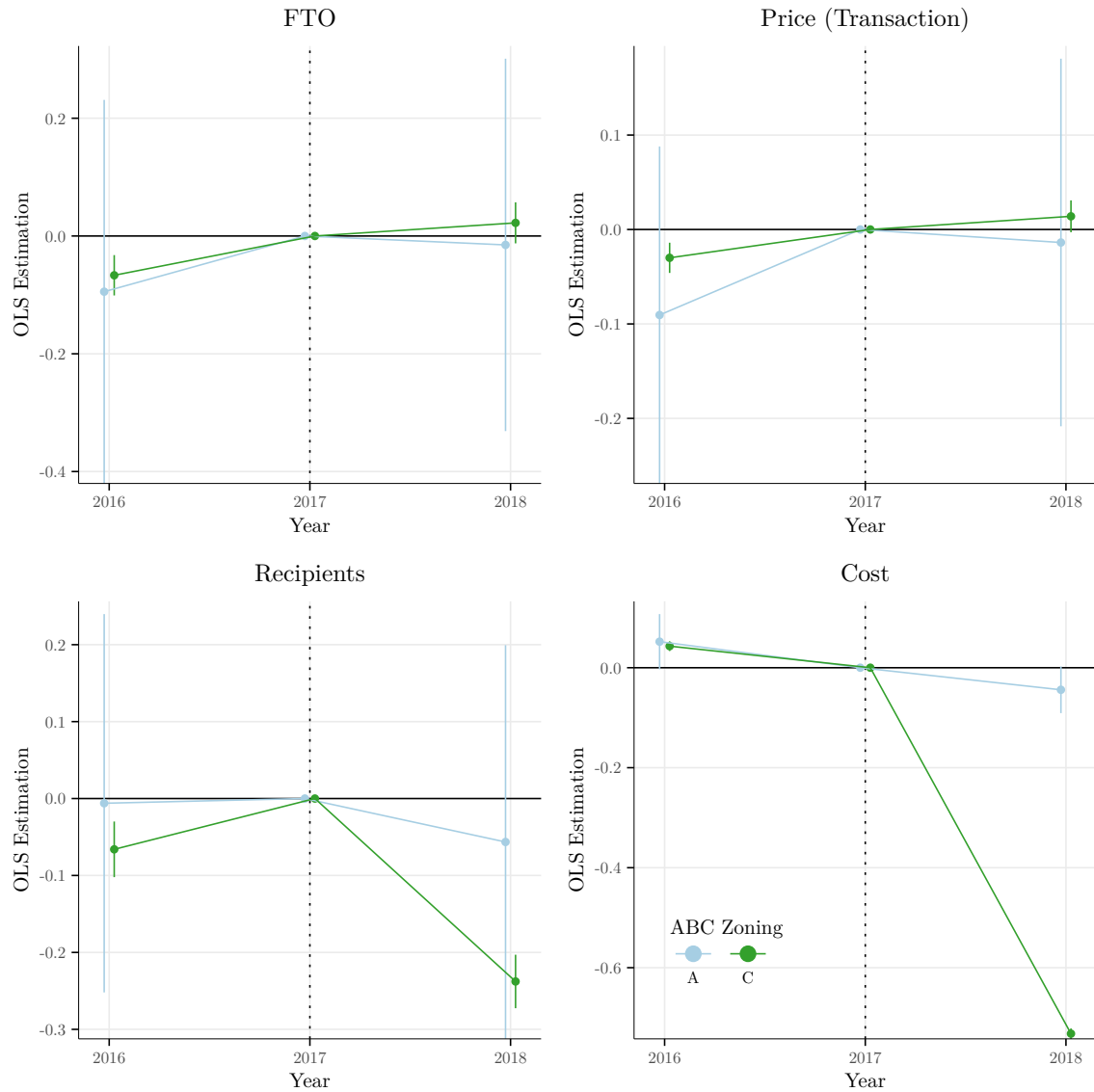


Figure D.5. OLS Results from DiD Identification Strategy restricted to A and C Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to A and C municipalities. C municipalities experience a cut in the coverage share that occurred in January 2018. Conversely, A municipalities do not experience any cut in the covering share. Our unit of observation is the municipality. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D.6 DiD Estimation for A and B₁ Areas

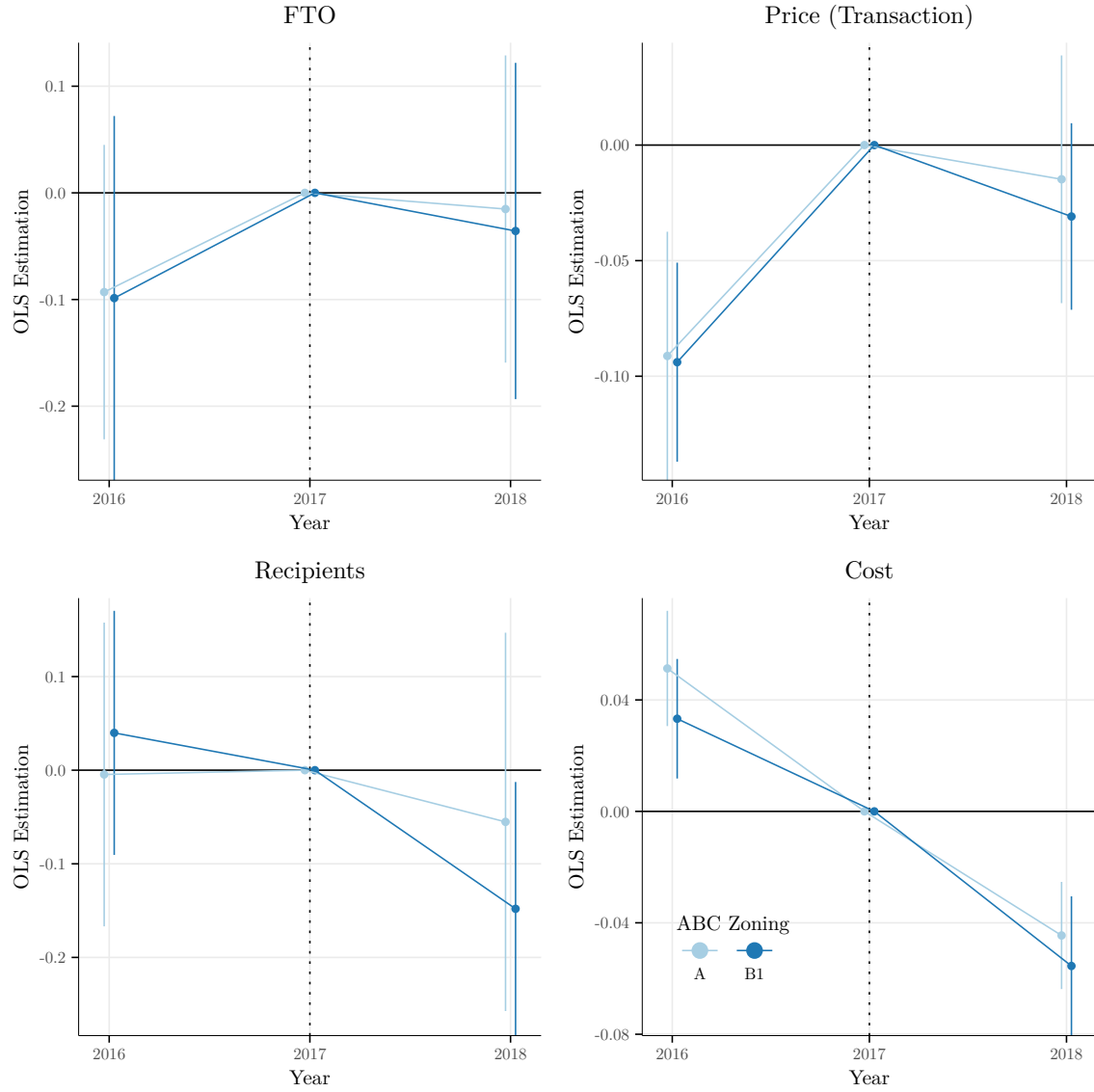


Figure D.6. OLS Results from DiD Identification Strategy restricted to A and B₁ Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to A and B₁ municipalities. Both groups of municipality experience no cut in the covering share. Consequently, it corresponds to a placebo test. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.

D.7 DiD Estimation for B₂ and C Areas

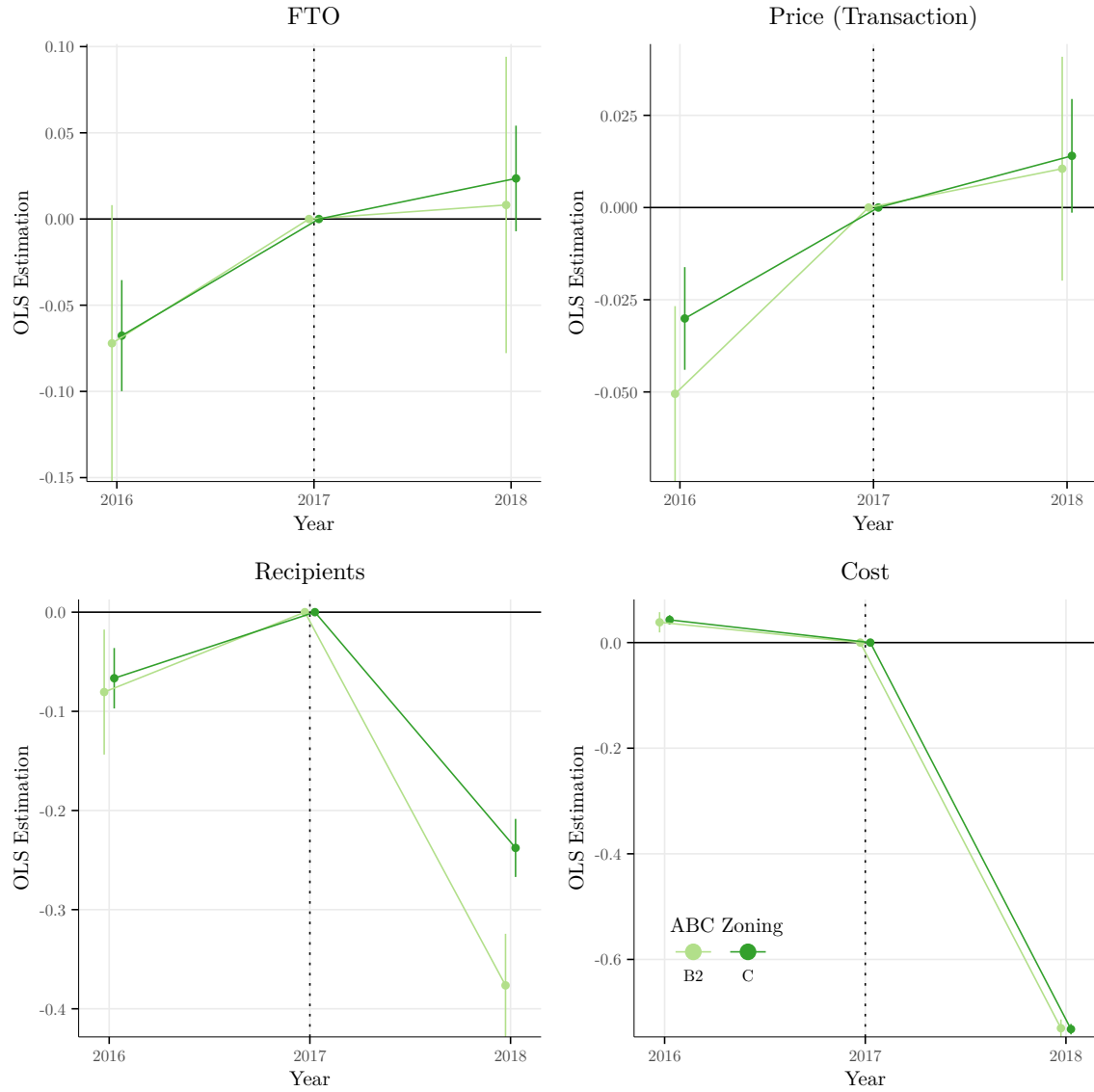


Figure D.7. OLS Results from DiD Identification Strategy restricted to B₂ and C Areas

Notes: We represent results from event study for four outcomes of interest (number of first-time owners, average transaction price, number of recipients and policy cost). We restrict samples to B₂ and C municipalities. Both groups of municipality experience a cut in the covering share. Considering potential heterogeneity in treatment effect, it cannot be considered as a placebo test. We report standard error in vertical line with a 95% confidence interval. Standard errors are clustered to the commuting zone.

Sources: Authors' Calculation based on SGFGAS, DV3F, *Fichiers Fonciers* and INSEE data.